Let's see, what you have learnt 9.2

I. What items of waste are harmful for the soil?

_____

II. Give the name of some organic manure?

_____

III. What can be used instead of chemicals pesticides to protect the soil from pests and insects?

_____

IV. What type of farming is beneficial for the soil, for humans and for animals and birds?

_____

Let's Revise

- Often due to the flow of water or strong winds, soil particles break and settle at another place. This breaking of the soil is called soil erosion.
- Due to erosion, the fertile top layer of the soil is destroyed. This decreases the crop yield.
- Soil erosion can be prevented by making boundaries around the farms, terrace farming, growing trees, plants, grass and shrubs, building check dams and making provision for proper outflow of water.
- The soil is polluted when it mixes with harmful and poisonous substances and elements. When the soil deteriorates it is called soil pollution.
- The cause of soil pollution is polythene, pieces of glass and iron and poisonous waste are mixed in soil.
- Due to soil pollution the soil weakens and it affects its fertility. This leads to loss of yield.
- To prevent soil pollution we should not throw harmful garbage in the soil or in farms. We should adopt organic farming.

Exercise

I. Read and answer the questions.

Organic farming is a boon to the soil, environment and to our health. Chemical fertilizers and pesticides are harming us. Therefore instead of chemical fertilizers we should use organic manure as much as possible because it is beneficial. Cow-dung manure, compost, vermicompost, khali manure and green manure are the main organic manures.

i. For whom is organic farming a boon?

_____

ii. What is harming us?

_____

iii. The use of which product is beneficial in place of chemical fertilizers.

_____

iv. Name any two organic manures?

_____

II. Answer the following questions based on the lesson:

i. What is gully erosion?

_____

ii. Which danger increases when the eroded soil gets deposited in the river?

_____

iii. What happens if we sprinkle cow-urine on the crops?

_____

III. Match the Following by drawing lines:

• Green	Farm
• Soil	Water
• Crops	Garbage
• Rainfall	Khali
• Steps	Erosion
• Neem	Manure
• Waste	Harvest

IV. Choose the right word and fill in the blanks.

i. Rapid flow of water creates small - small ..... (rings / gullys)

ii. Making a boundary is also called ..... (Dolbandi / Golbandi).

iii. We can prevent ..... by making our farms into small terraces. (Ill feelings /Erosion)

iv. Green manure is a ..... for farms (boon / danger)

V. Put a tick (✓) on the right answers and a (✗) on the wrong answer.

i. When the water flows rapidly it takes with it small particles of the soil. ( )

ii. The top layer of the soil is most useless. ( )

iii. It is beneficial for everyone to cut and destroy forests. ( )

iv. The roots of the trees hold the soil. ( )

VI. How is the farm soil polluted?

_____

_____

_____

_____

_____

_____

Come, let's do

- There must be parks and garden nearby. Some empty land where there is lots of grass and also where there is no grass. There can be land which is on a slope and also flat surfaces. Go there after it has rained. Look at the effects of rain water at different places. Can you see the difference? Think, why there is this difference?
- Before putting manure made from cow-dung, sieve it. After sieving it, see what things are left. Think, whether it is right to put these left over things in the farm soil. What can we do to check that these things are not found in the cow-dung? Ponder over it and then follow it.

ANSWERS

Let's see, what you have learnt

9.1 I. Breaking away of soil.

II. Due to rain and wind.

III. By doing terrace farming.

IV. The roots of the trees and plants hold the soil.

9.2 I. Polythene and pieces of glass and iron

II. Compost, Cow-dung manure, Vermicompost, Green manure and Khali manure.

III. Cow-urine, neem, tulsi, dhatura, tobacco, garlic and matka

IV. Organic farming.

Exercise

I. i. Environment, Soil and for our health.

ii. Chemical fertilizers and pesticides.

iii. Organic manure.

iv. Cow-dung manure or Vermicompost.

II. i. Rapid Flow of water.

ii. Of floods.

iii. Mixing of poisonous items.

iv. Protection from pests and diseases

III. i. Green Manure

ii. Soil erosion

iii. Yield from Crops

iv. Rain water

v. Step farming

vi. Neem khali

vii. Waste and garbage.

IV. i. Gullys

ii. Dolbandi

iii. Erosion

iv. Improvement

v. Boon

V. i. ✓

ii. ✗

iii. ✗

iv. ✓

Assessment Sheet 3

From Lesson 7 to 9

1. Fill in the blanks

a. All animals are dependent on ..... for food.

b. Plants make their food through the process of .....

c. Those creatures that can eat both plants and animals are called .....

d. Those who feed on dead bodies are called .....

e. ..... makes the soil soft and fertile.

f. It is necessary to make the seeds ..... and ..... before sowing.

g. Grow grass, shrubs and ..... on empty land.

h. Polythene, glass and iron should not be ..... in farm soil.

i. Use ..... neem ..... and rotting leaves.

j. Green grass is a ..... for farms.

2. Write the answer to the questions

a. How do plants make their food?

_____

b. What is the process of making food called?

_____

c. Who is an omnivorous creature?

_____

d. How is the soil made soft and fertile?

_____

e. What is grown in sandy soil?

_____

f. What is another name for acidic soil?

_____

g. Which medicines are used for treating seeds?

_____

h. How many types of crops? Name them.

_____

i. What are the reasons for soil erosion?

_____

j. What are the effects of soil erosion?

_____

k. Name any two methods to prevent soil erosion?

_____

l. Give the names of organic manure.

_____

10

Sources of Energy and their Importance



We will learn in this lesson

- Why is energy required?
- Different sources of energy and kinds of energy.
- What are the alternatives of obtaining energy?
- How can energy be saved?

You must have seen that if you do not eat food for a long time, you start feeling weak. Just think why do you feel weak? This means that there is some energy or power in our body. This strength helps us in conducting our day to day work. When this energy is over, we start feeling tired. We feel hungry and need food.

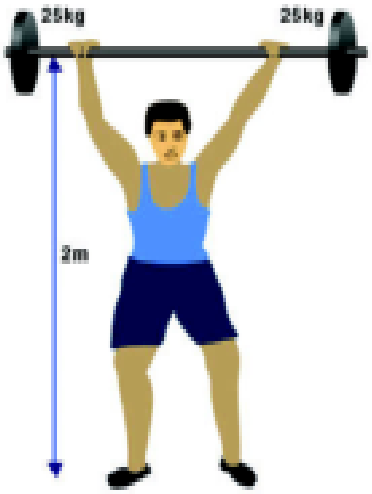
Similarly if there is no petrol or diesel in the car it does not run. From this we know that to do some work some energy or power is needed. In scientific language we call this strength, 'energy'. Energy is necessary for us to do work in our daily life. Energy is obtained from many sources and things.

In this chapter we will learn about the requirement of energy in our daily life, different sources of energy and alternatives of obtaining energy.

10.1 Why is energy required?

You have learnt that energy is used to do some work. Do you know that we require energy even when we are sleeping? Because when we are sleeping our body organs like lungs, heart etc. keep working. In our daily life, we require energy to do all our work. For example:

- for lifting weight
- for washing clothes
- for cooking food
- for running around



Lifting weight



Cooking food



Washing clothes

Now you must have understood that energy is required to do every work in life, even for eating, passing stool, urine etc. Come, now let's discuss from where and in what form we obtain energy.

10.1.1 Sources of Energy

We obtain energy from different things. These things are called sources of energy. Energy obtained from these sources is of different varieties. These are called kinds of energy. There are mainly two sources of energy.

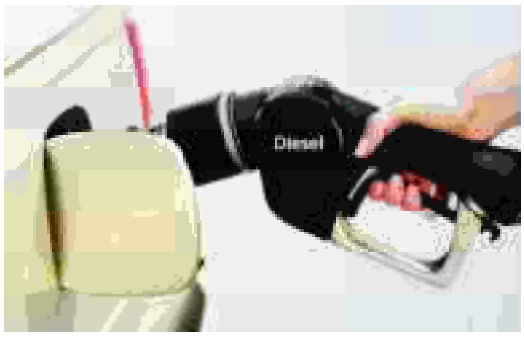
1. Non-renewable sources of energy

In most of the houses people use coal, kerosene oil, cooking gas, petrol, diesel etc. to obtain energy. These sources of energy finish after being used once. Therefore they are called non-renewable sources of energy. Just think, if we keep using them continuously, won't they come to an end or finish?

- Coal
- Kerosene oil
- Diesel
- Cooking Gas









Sources of non-renewable energy

Do you know?

If we see the current and future requirement of energy, it can be assumed that our oil and gas reserves will be finished in 30-35 years. Coal reserves can last up to 100 years.

2. Renewable sources of energy

You have read that the sources from which we get energy for our day to day work will gradually and slowly finish. But we get energy mainly from food. Nature provides food for us forever. Apart from this, there are also some sources from which we always keep getting energy and they never finish. These are - wind, water, sunlight. These sources of energy are known as renewable sources of energy.

Let's see what you have learnt 10.1

I. What is the power to do work called?

_____

II. Why do we feel tired after doing excessive work?

_____

III. Why do we require energy?

_____

IV. How many kinds of sources of energy are there? Give two examples of each.

_____

10.2 Kinds of Energy

You have learnt about the different sources of energy. Come let's learn about the different kinds of energy that we get from these sources.

10.2.1 Energy received from food

There are many kinds of flora and fauna in our environment. We use them in the form of our food. This food provides us energy for doing our day to day work. For all living beings, plants and trees are the main source of energy. Plants and trees make their own food from sunlight. Energy received from food is called food energy.



Sources of food energy

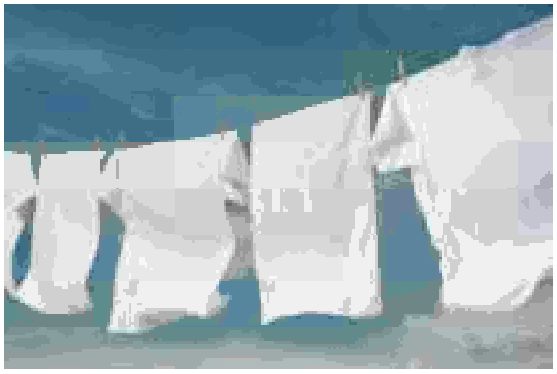
10.2.2 Solar Energy

We see that the sun always provides energy for all living beings from its light. Energy received from the Sun is called solar energy. This energy is obtained in the form of light and heat. Solar energy is used to dry clothes, food grains and by plants to make their food. These days this energy

is also used for cooking food, heating water, lighting bulbs etc. From solar energy, we get Vitamin D which strengthens our bones. Solar energy is also used for running small vehicles. The Sun always keeps giving energy, and this energy never decreases. Therefore the Sun is known as a source of energy. Because of this energy, life is possible on earth. Our crops grow because of sunlight. To use solar energy, solar cookers, solar water heaters, solar cells etc. are available in the market.



Solar Cooker





Drying clothes, drying food grains

10.2.3 Wind Energy

Have you ever thought why a pinwheel spins? If children run with a pinwheel and blow air on its blades, then it starts spinning faster. Why does this happen? When the blades of the pinwheel get excessive air then in the blades the work of spinning takes place. So we can say that this work gets done due to wind energy. Therefore in places where strong winds blow for most of the year, wind mills are used for taking out water from the wells and for producing electricity. Wind mills are run by energy produced by wind. Wind mills are also used to grind flour.



pinwheel

10.2.4 Hydro Energy (energy obtained from water)

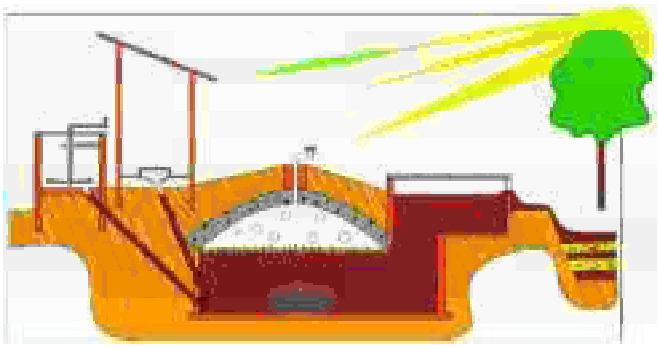


Hydro Power Stations

You must have seen that big wooden logs and trees are floated in rivers to shift them from one place to another. This means that flowing water just like flowing wind, also produces energy. Therefore in hydro power stations electricity is produced from flowing water.

10.2.5 Organic Energy (Energy obtained from faeces-urine, dead bodies and dry plants)

You must have heard the name of gobar gas. Think how it is produced. Actually, animal dung, dead animals, dry leaves, branches and useless residues of crops are used to produce organic energy. These substances are put into a biogas plant to produce gas. This gas is used for cooking food and producing light. The residual substance in the biogas plant is used as manure.



Biogas Plant

10.2.6 Electrical Energy (Energy received from electricity)

We get light on lighting an electric bulb, but to light the bulb electricity is used. This is called electrical energy. This energy is used to run small machines and big factories, for example fan, tubewell, radio, Television, computer, fridge etc. These days' trains are also run with electrical energy.



Fan, other appliances run with electricity

10.2.7 Magnetic Energy (Energy received from magnets)

We all know that magnet attracts iron towards itself. This power of magnet is called magnetic energy. It is used to separate iron from stock piles of scrap. This energy is also used in factories to lift heavy iron products. Apart from this, electrical appliances also use magnetic energy, for example- electric bell.



Crane separating iron, electric bell

10.2.8 Alternative sources of energy

We use coal, kerosene oil, petrol etc. to generate energy. These things are slowly decreasing. Because of excessive use, these can finish soon. Therefore we must make greater use of other sources of energy, like solar, wind, water, organic etc. These sources are also called alternative sources of energy. You have already learned about them. We must use more of solar cookers, wind mills, electricity and biogas extensively.



Cooking food from Solar Cooker

We generally use coal, kerosene, petrol etc. for obtaining energy. These objects are depleting gradually. Their excessive use may deplete them faster. Therefore, we should use other sources of energy such as sunlight, wind, water and organic objects. These are called alternative sources of energy also. You have already read about them. We should use more of bio-gas, electricity, solar energy, wind mill etc. mainly.

Let's see, what you have learnt 10.2

I. Write the names of seven types of energy.

_____

_____

II. Fill in the blanks:

i. On earth life is possible because of ..... energy.

ii. In hydro power stations ..... energy is used.

iii. In bio gas plant ..... energy is produced.

iv. To run a fan ..... energy is used.

v. Magnetic energy is used to separate ..... from a stockpile of scrap.

10.3 How can energy be saved?

There are some sources of energy which are gradually decreasing. A time may come when they are completely finished, for example coal, petrol, diesel, natural gas etc. Therefore it is necessary that we save energy. Energy may be obtained from any source, but its saving will prevent an energy crisis. Today the Government has also provided alternatives to energy. To save energy we must make efforts from our homes. If we keep the following points in mind while working, we will be able to save energy.

- If it is not needed then at home and at places of work, the switches of bulbs, tubes, fans etc. should be switched off.
- Do not leave water taps open. Because to make water available to us, energy is spent.
- While cooking food, utensils should be covered.
- To cook pulses, vegetables etc. do not put more water than the required. Put only that much water it is needed.
- Before putting pulses and rice on the stove, soak them in water for some time.
- Use public transport instead of personal vehicles.
- Use fuel efficient appliances. For example use CFL bulbs or tube lights instead of normal bulbs. Use pressure cookers instead of open utensils to cook food.

Let's see, what you have learnt 10.3

I. How can alternative sources of energy be used? Mention two ways.

_____

_____

II. Mention four ways in which energy can be saved.

_____

_____

_____

_____

Let's Revise

- The power or capacity to work is called energy.
- Energy is necessary for every work in our life.
- There are two types of sources of energy: Non renewable and Renewable.
- Coal, petrol, diesel, kerosene oil etc. are non renewable sources of energy.
- Food, sunlight, wind, water and biogas (organic) are renewable sources of energy.
- The sun is considered as the inexhaustible source of energy on the earth.
- There are many kinds of energy - food energy, solar energy, wind energy, hydro energy and biogas energy.
- The alternatives to obtain energy are - sun, wind, water, biomass.
- Energy can be saved by wisely using it according to our requirement.

Exercise

I. Fill in the Blanks:

i. Energy is the ..... to do work.

ii. We need energy for ..... work in our life.

ii. The sources of energy are of ..... types.

iv. ..... are non renewable sources of energy.

v. ..... are renewable sources of energy.

II. How many types of energy are there? Write their names with examples.

_____

_____

III. Give reasons for the following:

i. Why do we become weak if we do not get food?

_____

ii. Why does a pinwheel spin faster when we blow air on it?

_____

iii. In a hydro electric station, how is energy generated?

_____

_____

iv. Why are biomass plants made?

_____

_____

IV. What are the alternatives of obtaining energy?

_____

_____

_____

_____

_____

V. Suggest five ways of saving energy.

_____

_____

_____

_____

_____

_____

ANSWERS

Let's see what you have learnt

10.1 I. Energy

II. Due to lack of energy

III. to do all work

IV. Two types -

i. Non Renewable - Coal, petrol

ii. Renewable - Sun, wind

10.2 I. Food energy, solar energy, wind energy, hydro energy, organic energy, electrical energy and magnetic energy

II. i. Solar ii. water

iii. organic iv. electrical

v. iron

10.3 I. Answer yourself

II. Answer yourself

Exercise

I. i. capacity ii. every

iii. two iv. coal and kerosene oil

v. sun and wind

II. See Answer No. 10.2.1

III. i. due to lack of energy

ii. due to wind energy

iii. with water

iv. to generate organic energy from dung and waste products

IV.. Sun, water, wind

Come, let's do and observe

To make a solar cooker take a box made of wood or thick card board. Stick black paper on the outside and inside of the box. Cover the mouth of the box with a glass. Now put a mirror on the inside of the cover of the box. Open the cover and put it on top of the box in such a way that sun rays fall on the mirror and then reach the glass and the inside of the box. Inside the box put bowls with rice, pulses and vegetables soaked in water. After some time see how the Sun God prepares food for you. You can also use aluminum foil instead of a mirror.

11

Solar System and Earth



We will learn in this lesson

- Names of the different planets of the solar system and our planet- Earth
- Difference between planets and satellites and different phases of the moon
- Natural and manmade satellites
- Locating places on earth- lines of Latitude and Longitude
- Reason for different times at different places

From ancient times man has taken interest in events happening around him. He has constantly tried to understand the sky, the sun, the moon, stars, day-night, increase and decrease of the moon, change in seasons etc.

In India, from ancient times, learned men made cycles of the zodiac signs on the basis of different groups of stars. They made the Indian calendar and dates on the basis of the phases of the moon. Even today horoscopes are made on the basis of the zodiac signs.

Have you ever thought about how time is estimated? Why the time is different at different places on the earth?

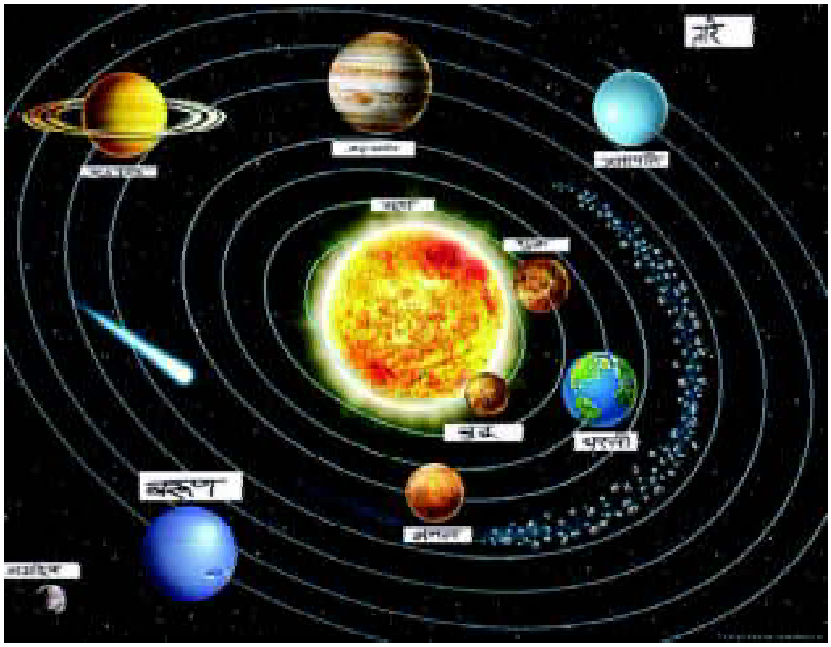
Come; in this lesson let's learn about the earth, the sky and time.

11.1 Our Solar System and its Planets

Every day you see the sun, the moon and stars in the sky. All these celestial bodies are a part of our huge universe. There are countless stars in the universe. The Sun is also a star.

The Sun is the star closest to the earth. This is why it looks bigger than other stars. Many big and small celestial bodies keep revolving around the sun. From amongst these, eight big bodies are called planets. Their names are- Mercury, Venus, Earth, Mars, Jupiter, Saturn, Neptune and Uranus.

Planets are celestial bodies which go around the sun.

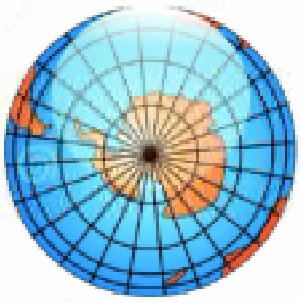


Solar System

Our earth is a planet of the sun. This is the only planet to have life on it. This is possible because of the presence of air and water on the earth. Our earth is geoid/spherical. It has two poles - North Pole and South Pole. It revolves around the sun in a fixed path called orbit.

The earth is tilted slightly to one side on its axis. The earth moves on its axis. This movement of the earth on its axis is called rotation. The earth takes 24 hours to complete one rotation on its axis.

The movement of the earth around the sun is called revolution. The earth takes 365 days and 6 hours to complete one revolution around the sun.



North Pole



South Pole

Let's see, what you have learnt 11.1

I. What are planets?

_____

II. Fill in the blanks:

i. The movement of the earth on its axis is called .....

ii. To complete one rotation the earth takes .....

iii. The movement of the earth around the sun is called .....

iv. The earth takes ..... to complete one revolution around the sun.

III. Write the names of the eight planets of the sun:

_____

_____

11.2 Planets and Satellites

We know that the celestial bodies which move around stars are called planets. In the same way the celestial bodies which move around planets are called satellites.

The moon is a satellite of the earth. It moves around the earth.

11.2.1 Phases of the moon and estimating dates on the basis of these phases

You know that the moon moves around the earth. This is the reason why sometimes the moon is visible to us and sometimes it is not. The moon takes 29 ½ days to complete one revolution around the earth. The moon has no light of its own. Just like the earth it also gets its light from the sun. One half of the moon is always illuminated by the sun while the half not facing the sun is in darkness. Because the moon moves around the earth its lit up part is always changing. This change is the reason why we see a different phase of the moon every night. Sometimes it is round like a plate and sometimes incomplete like a sickle. Its shape keeps increasing or decreasing.

The night when the moon shines brightest in the sky is called the full moon night or purnima. After the full moon night, the bright part of the moon decreases gradually every night and one night it is completely hidden. This means we cannot see it in the sky. This is called the new moon night or

amavasya. After the new moon night the moon starts increasing gradually every night. It appears fully shining again on fifteenth night. The increase and decrease of the lit part of the moon is called the phases of the moon.

In our country counting of dates is done on the basis of the phases of the moon. The Indian calendar, made on this basis, has 2 phases (Paksha) of the month and each phase of the moon lasts for 15 days. The fortnight between Full Moon and New Moon is called Krishna Paksha. And the fortnight between New Moon (Amavasya) to Full Moon (Purnima) is called Shukla Paksha. On this basis the dates from first to the fourteenth are estimated. The monthly calendar is also based on this.

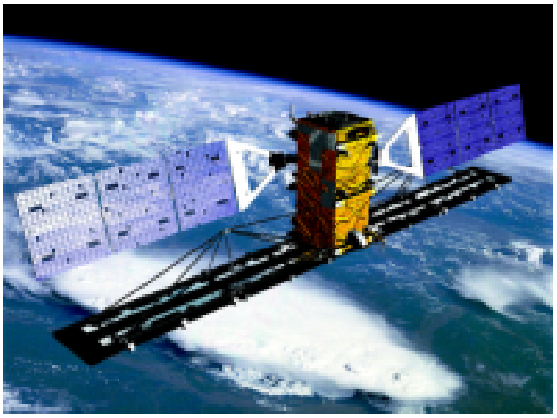


Phases of the Moon

11.2.2 Natural and Artificial Satellites

The celestial bodies moving around the planets are known as natural satellites. Man made machines which orbit around planets are known as artificial satellites.

Man has made many satellites of the earth. These satellites keep orbiting the earth and taking images of the earth. Weather forecast for coming days is based on these images. These satellites help in giving information about any terrible forest fire.



Artificial Satellites

Some artificial satellites have also been sent into space, which help in sending information from one corner of the world to another. It is through them that sitting in India we get to know what is happening in America through the medium of television. A cricket match being played in Africa is telecast to all corners of India. All this is possible only because of satellites.

Let's see, what you have learnt 11.2

I. Put a correct (✓) or incorrect (✗) sign.

i. The celestial bodies that move around stars are called satellites. ( )

ii. Moon is a satellite of the earth. ( )

iii. When the moon shines fully in the night sky it is called amavasya. ( )

iv. The increase and decrease in the size of the moon is called phases of the moon. ( )

II. Fill in the blanks:

i. Man made machines which orbit around planets are known as ..... satellites.

ii. Through artificial satellites sitting at home we get ..... about the world.

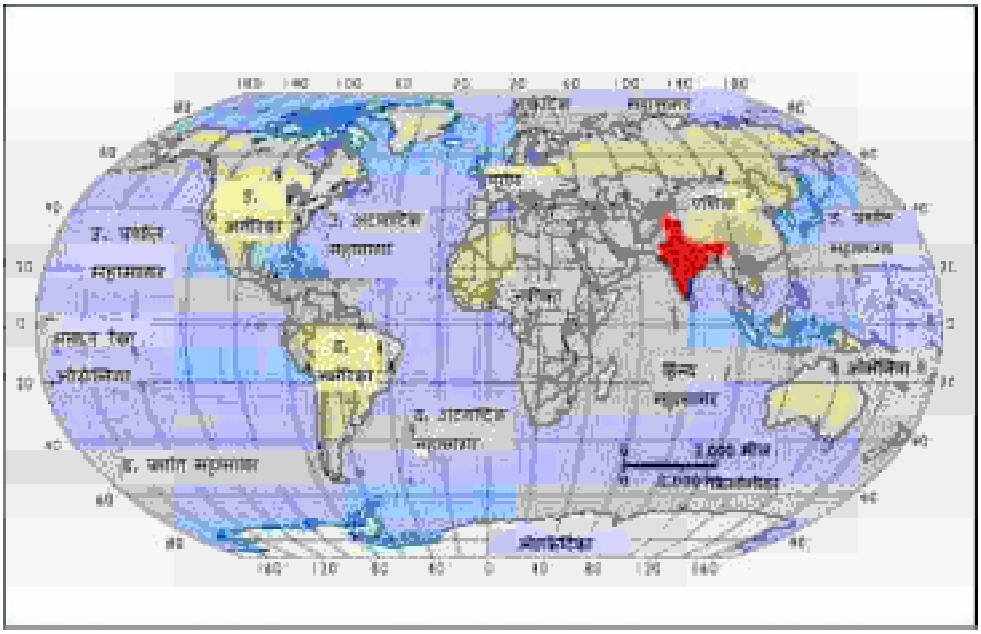
iii. The moon takes ..... days to complete one circle around the earth.

11.3 Latitudinal and Longitudinal Lines

The size of the earth is huge. There are about 196 countries on the earth. Every country has many states, cities and towns. Under these circumstances it is essential to have some method of locating places on the earth easily. This is the reason why many imaginary lines have been drawn on the earth. These lines divide the earth into many blocks and help in locating places. These lines are called the Lines of Latitude and Longitude. Come, let us know more about these imaginary lines.

11.3.1 Latitudinal Lines

Scientists have made a model of the earth. This is called a globe. Another impression of the globe is printed on paper. This is called a map. On the globe and map, imaginary lines have been drawn from East to West and from North to South.



India and the world

There are 180 imaginary lines from East to West. These are called latitudes. Similarly imaginary lines drawn from North to South are called Longitudes. These are 360 in number.

The central most line (latitude) is called the equator. This line moves through the centre of the earth, hence it is called equator. The Northern part is called the Northern hemisphere and the southern part is called the Southern hemisphere. Similarly the central most line (longitude) is called the Meridian line. The part to the East of the meridian is called the Eastern hemisphere and the part to the west of the Meridian is called the Western hemisphere.

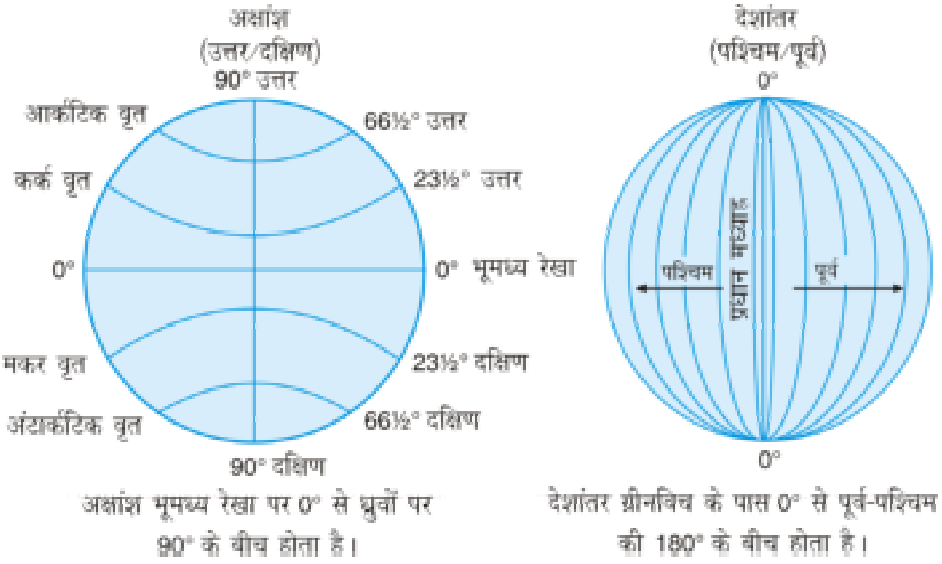
Other important lines are as follows -

Tropic of Cancer - To the North of the equator is the Tropic of Cancer at 23 1/2°.

Tropic of Capricorn - To the South is the tropic of Capricorn at 23 1/2°.

Arctic Circle - This is to the North of the equator at 66 1/2°.

Antarctic Circle - This is to the South of the equator at 66 1/2°.



Lines of Latitude and Longitude

11.3.2 Longitudinal Lines

On earth the imaginary vertical lines are called lines of longitude/longitudinal lines.

You must have seen vertical lines in the middle of a peeled orange. Longitude lines look just like them. The distance between these lines is more near the centre but decreases as we move towards the poles. Longitude lines meet at a point on the Poles. The Meridian line (0° Longitude) divides the earth into two equal parts - Eastern Hemisphere and Western Hemisphere. There are a total of 360 Longitude lines.

11.3.3 Different times at different places

In Longitude lines, time is measured from 0° Longitude. There is a difference of 4 minutes in between two Longitude lines. 0° Longitude line passes through Greenwich in England. The time of all countries is determined according to the time at Greenwich. India is located between 68°7' East Longitude and 97°25' East Longitude. But there is the same (i.e. one) Standard time in the whole India. If the same Standard time had not been adopted then there would be different times at different places. In India the time at 82 1/2° East Longitude (which passes through Allahabad) is taken as the Standard time. Similarly the time of all countries is determined.



Let's see, what you have learnt 11.3

Fill in the blanks:

i. The model of the earth is called a .....

ii. Imaginary Latitude lines have been drawn from ..... to .....

iii. The number of longitude lines is .....

iv. The earth is divided into two equal parts by the .....

v. The Tropic of Cancer is ..... North of the equator.

vi. Tropic of Capricorn is ..... South of the equator.

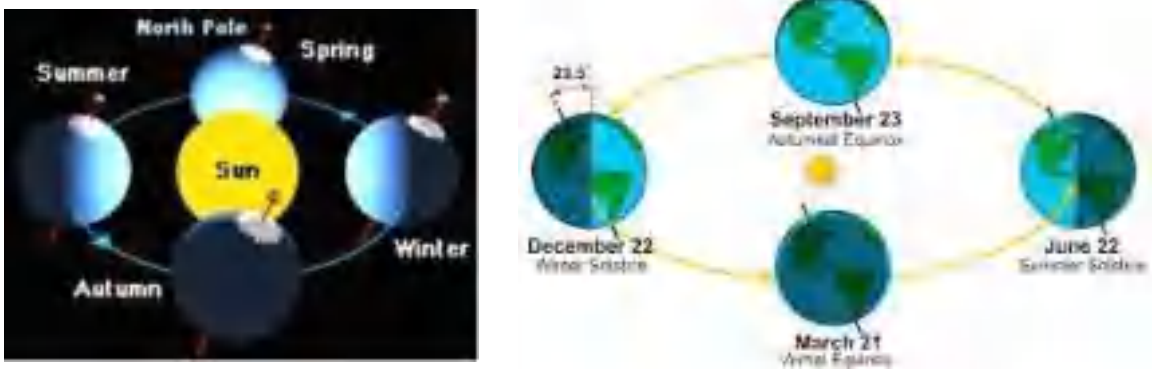
11.4 Day and Night becoming long and short

The earth is tilted on its axis toward one side. Therefore sunlight does not fall on the earth evenly. For this reason the days and nights are not equal at all places. While the earth is revolving around the sun, sometimes days are long and nights are short in the Northern Hemisphere and sometimes in the southern Hemisphere.

Apart from the equator, at all places there is a difference in the length of days and nights. Days and nights are equal at all times on the Equator.

Because the earth is tilted on its axis, on both the Poles there is night for 6 months and day for 6 months.

The rays of the sun fall in a slanting manner on the Poles. Therefore it is always cold there.



Day and Night being long and short

Position on 21st June - On 21st June the rays of the sun fall directly on the Tropic of Cancer. Because of the direct sun rays, at this time the days are the longest and hottest in the Northern Hemisphere and the nights are the shortest. Conversely the Southern Hemisphere is cold at this time. 21st June is the longest day in the Northern Hemisphere.

Position on 22nd December - At this time the rays of the sun fall directly on the Tropic of Capricorn. At this time, the days are long and nights short in the Southern Hemisphere. Conversely in the Northern Hemisphere days are short and nights are long. In the Northern Hemisphere it is cold at this time. 22nd December is the shortest day.

Position on 21st March and 23rd September - During these days the rays of the sun fall directly on the Equator. Due to this, at all places days and nights are of equal length and the season is the same. For this reason both these positions are called Equinox or days and nights are of equal length. The position of 21st March is called Spring Equinox and the position of 23rd September is called Autumn Equinox.

Let's see what you have learnt 11.4

Fill in the blanks:

i. The earth is ..... on its axis.

ii. On the Poles there is night for 6 months and day for .....

iii. The rays of the sun fall in a ..... on the Poles.

iv. On 21st June the rays of the sun fall straight on .....

v. On 21st March and 23rd September the rays of the sun fall directly on the .....

Let's Revise

- The Sun is a star.
- Celestial bodies revolving around any star are called planets.
- The sun has eight planets. Our earth is also a planet.
- The earth is not completely round, it is Geoid/Spherical.
- In the Northern direction there is North Pole and in the Southern direction there is South Pole.
- Besides rotating on its Axis, the earth also revolves around the sun.
- The moon is a satellite of the earth.
- The increasing and decreasing of the lit part of the moon is called the phases of the moon.
- On the basis of the phases of the moon, horoscopes or cycles of signs of zodiac are prepared and dates are counted.
- Man made machines which orbit around planets are known as artificial satellites.
- The movement of the earth around its Axis is called Rotation.
- The Earth takes 24 hours to rotate once on its Axis.
- The movement of the Earth around the Sun is called Revolution.
- The earth takes 365 days and 6 hours to revolve round the sun.
- When the Earth is close to the sun it is hot and when it is away from the sun it is cold.
- Increasing and decreasing distances of the earth from the Sun causes increase and decrease in temperature. This also makes seasons.
- When the earth is closest to the sun it is called perihelion. This occurs on 3rd January.
- When the earth is farthest from the sun it is called aphelion. This occurs on 4th July.
- From 21st March to 23rd September days are long and nights are short in the northern hemisphere.
- From 23rd September to 21st March the sun remains for more time in the Southern Hemisphere as a result of which days are long and nights are short here.
- Because the earth is tilted on its axis, on both the Poles there is night for 6 months and day for 6 months.

- The rays of the sun fall in a slanting manner on the Poles. Therefore it is always cold there and the Poles are covered with snow.
- The model of the earth is called a Globe.
- When the Globe is printed on paper, a map is prepared.
- There are 180 lines of Latitude and 360 lines of Longitude.
- There is a difference of 4 minutes in between two Longitude lines.
- In Latitude lines the Equator lies at 0° and it divides the Earth into two parts.
- On 23 1/2° in the Northern Hemisphere lies the tropic of Cancer. On 23 1/2° in the Southern Hemisphere lies the tropic of Capricorn.
- In the Northern Hemisphere on 66 1/2° is the Arctic Circle. In the Southern Hemisphere on 66 1/2° is the Antarctic Circle.
- The Prime Longitude line goes through 0° Longitude at Greenwich in England. The time of all countries is determined from here.
- In our country the time based on 82 ½° E line of longitude (which passes through Allahabad) is taken as the Standard time.
- The Moon moves around the earth once in 29 ½ days.



Exercise

I. Fill in the Blanks:

i. The earth is a .....

ii. The shape of the earth is .....

iii. The moon is a ..... of the earth.

iv. The path of the earth's revolution is .....

v. In the world there are approximately ..... countries.

vi. Imaginary lines from East to West are called .....

vii. Imaginary lines from North to South are called .....

viii. The moon moves around the earth once in ..... days.

ix. The moon shines from the light of the .....

II. Answer the following Questions:

i. What are planets?

_____

ii. What is a revolution?

_____

iii. What are imaginary lines?

_____

iv. What is a globe?

_____

v. What is a map?

_____

vi. What is an artificial satellite?

_____

III. Match Column A with Column B

	Column A		**Column B**
i.	The sun has	a.	is moon
ii.	The earth's satellite	b.	in 29 days 6 hours
iii.	The earth takes one circle around the sun	c.	is 180
iv.	The moon takes one circle of the earth	d.	is 360
v.	The number of latitudes	e.	is on 0 Latitude
vi.	The number of longitudes	f.	8 planets
vii.	The Equator	g.	is on 23 1/2° in Northern Hemisphere
viii.	The Tropic of Cancer	h.	in 365 days 6 hours
ix.	The Tropic of Capricorn	i.	is on 23 1/2° in Southern Hemisphere

IV. Write the answer in one word:

i. What is prepared from zodiac cycle?

_____

ii. In how many days does the earth revolve around the sun?

_____

iii. From where is the time in India determined?

_____

iv. From where does the 0° Longitude pass?

_____

v. What is the shape of the earth?

_____

vi. On earth what are the imaginary lines drawn from one Pole to another Pole called?

_____

ANSWERS

Let's see, what you have learnt

11.1 I. Celestial bodies which move around the sun.

II. i. rotation ii. 24 hours

iii. revolution iv. 365 days and 6 hours

III. Mercury, Venus, Earth, Mars, Jupiter, Saturn, Neptune, Uranus

11.2 I. i. ✓ ii. ✓ iii. ✗ iv. ✓

II. i. artificial ii. information iii. 29 ½ days

11.3 i. globe ii. North, South

iii. 360 iv. Equator

v. 23 1/2° vi. 23 1/2°

11.4 i. oval ii. six months

iii. slanting iv. Tropic of Cancer

v. Equator

Exercise

I. i. planet ii. oval

iii. satellite iv. oval

v. 196 vi. Longitude

vii. latitude viii. 29 ½ days

ix. sun

II. i. Celestial bodies which move around the sun.

ii. The moving of the earth around the sun.

iii. Latitudes and longitudes.

iv. Model of the earth.

v. Globe printed on paper.

vi. Manmade machines which orbit around planets.

III. i. f vi. d

ii. a vii. e

iii. h viii. g

iv. b ix. i

v. c

IV. i. horoscope ii. 365 days 6 hours

iii. 180° Latitude iv. Greenwich

v. Geoid, Spherical vi. Longitude lines

12

Food and Health



We will learn in this lesson

- Detection of diseases due to deficiency of nutrients, protection from them and their cure.
- When more attention should be paid on eating.
- Which are the cheap and beneficial food items?
- How to store food items.

Whether food is eaten for taste or to fill the stomach, food is necessary for everyone. Food is necessary for the development of the body, for hard work and for protection from diseases. All this work is done by nutrients present in food. Deficiency of these nutrients can cause many diseases in the body. In this lesson we will learn about detection of such diseases, protection from them and their treatment. We will also learn when some people need to pay special attention to food, how to make cheap nutritious food and how to store food so that it does not become stale.

12.1 Diseases due to lack of nutrients

Food has the following nutrients:

- Carbohydrates and fats - these provide strength to the body

- Proteins - These help the body to grow
- Vitamins and mineral salts - These protect the body from diseases

Due to the deficiency of these elements many diseases can occur, like:

12.1.1 Anemia

In anemia a person's blood becomes thin. Generally the reason for this is lack of iron in the body. Symptoms of anemia are:

- Pale, transparent skin
- Yellowness in the inside of pupils of the eyes
- Yellow gums
- Whiteness in nails
- Weakness and tiredness
- If deficiency of blood increases, then there is swelling on the face and feet. Heartbeat increases. Breathing becomes faster. Nails become disfigured.

Prevention and treatment

- Eat food containing iron like bajra, ragi, jaggery, spinach, meat, fish, chicken and eggs.
- If the disease increases, then take iron tablets. Take help of the doctor.

12.1.2 Kwashiorkor (Wet Malnutrition)

Children get this disease due to protein deficiency.
Its symptoms are:

- Swollen face
- Body growth stops
- Wounds and scraped skin
- Swelling on hands and feet
- Loss of colour of skin and hair
- Thin and weak upper arms



Kwashiorkar Patient

Treatment

- Children suffering from Kwashiorkor need extra food. They should be given pulses, peas, legumes, soya bean, nuts, green leafy vegetables, milk, cheese, egg, meat and fish in greater quantity.

- Such children must eat food at least three times in a day. In between they should take light snacks.

12.1.3 Rickets (Dry Malnutrition)

Those children who do not get the right amount of food can get rickets. The symptoms of this disease are:

- Great loss of weight.
- Face becomes like that of an old person.
- Protruding stomach.
- Only skin shows on bones in the body.



Patient of Rickets

Treatment

Patient should be given nutritious food in large quantity.

12.1.4 Night blindness

Those people who do not eat food with Vitamin 'A' suffer from the disease of night blindness. Often children between the ages of 2 and 5 have this disease. Symptoms of this disease are:

- Patient cannot see clearly at night.
- In the beginning the eyes are dry, and then the white of the eye starts losing its colour. Slowly the patient becomes blind.

Prevention and treatment

- If possible the child should be breast fed for two years.
- Children above the age of six months should also be given other food containing Vitamin 'A', like: green leafy vegetables, yellow and red coloured fruits and vegetables, milk, egg. Liver has a sufficient amount of Vitamin 'A'.
- Give Vitamin 'A' capsules with doctor's advice.

Caution: Excess dosage of Vitamin 'A' is harmful.



Patient of night blindness

12.1.5 Rickets

Those children who do not get milk or whose body does not get sun light, can have crooked legs. Other deformities can also appear in the bones, like big joints, crooked bones, jugular bone loss.

Prevention and treatment

- To prevent and cure rickets the best remedy is sun light.
- Give milk and Vitamin 'D' to children. Vitamin 'D' is found in fish oil.

Some diseases of skin like dry and cracked skin, wounds on lips and corner of the mouth, bleeding of the gums.

All of these diseases are caused due to deficiency of fruits, vegetables and special vitamins.

Treatment

People suffering from these diseases should take food containing proteins and vitamins in large amounts, like green leafy vegetables, milk, meat, fish, eggs and fruits.

12.1.6 Goitre

Deficiency of iodine in food causes goitre. This disease occurs more in hilly areas. In this disease, there is swelling on the neck.



Goitre patient

Prevention and treatment

- Eat food with salt containing iodine.
- Old goitre can be removed by operation.

12.1.7 Mental slowness

If there is deficiency of iodine in a pregnant woman's food, then her child can be mentally challenged, dwarfed or deaf.

Prevention and treatment

- Always eat food with salt containing iodine.

12.1.8 Beri Beri

Deficiency of Vitamin 'B' causes a disease called Beri - Beri. In this there is severe pain in the muscles. There is lack of appetite. Food is digested with difficulty. Skin at the corner of the mouth gets cracked. The tongue becomes rough and red.

Let's see, what you have learnt 12.1

I. Write any two symptoms of anemia disease.

_____

_____

II. Match the diseases with their correct causes.

Disease	Cause
Kwashiorkor	Deficiency of Vitamin 'B'
Rickets	Deficiency of protein
Beri Beri	Deficiency of Vitamin 'A'
Night blindness	No sunlight on the body

III. Which two diseases are caused due to deficiency of iodine?

_____

_____

IV. Write the ways to prevent night blindness.

_____

_____

12.2 When should more attention to be paid on eating food

These people require special nutrients in food:

- Growing children specially need proteins. Therefore children above the age of six months should be given top food with milk, like mixture of rice and lentils, pulses, oats, vegetables, fruits etc.
- Special attention should be paid to the food of children during adolescence. Only then their bodies will develop properly. When menstruation begins girls should eat food containing more of iron. This will help in maintaining their blood level and there will be no deficiency.

- Pregnant and lactating women should eat bellyful of balanced food, because the child's health depends on the food that the mother eats. They should eat food containing calcium and iron.
- The body becomes weak during illness. In this condition many other diseases can be contracted. Therefore sick people should eat food which provides strength. Besides they should eat food which contains substances that prevent illness.



food for a pregnant woman

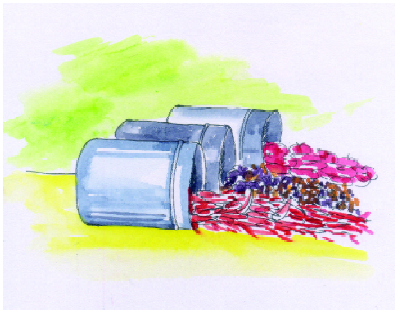
12.2.1 Inexpensive - nutritious food

There is nothing like nutritious and tasty food. Some people think that such food is expensive. But that is not so. With some intelligence nutritious and tasty food can be prepared. For example:



Mother breast-feeding her baby

- Mother's milk: This is the most easily available, healthy and complete food for a baby. If the mother eats food containing protein, then her milk will be the complete food for her child.
- Unpolished cereals: Unpolished rice and wheat bran flour contain more vitamins.
- Soak dry corn in some lime. The amount of vitamins and protein in corn will increase.
- Ragi and bajra contain iron and calcium in large amounts.
- Eating mixed cereals is better than eating only one type of cereal. This provides different types of protein to the body. Therefore mix many cereals and pulses and then cook them.
- Green beans, soya beans and other beans are the cheapest treasure of protein. If these are germinated, then the amount of protein and vitamins will further increase.



Soya bean

- Dark green leafy vegetables contain protein, iron and Vitamin 'A'.
- Cook rice and vegetables in less water. Cut vegetables just before cooking them. Do not overcook them. Do not throw away cooked rice water. This can destroy vitamins and mineral salts.

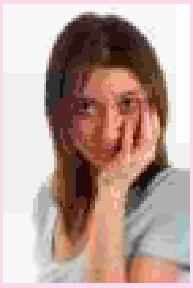
- Wash vegetables before cutting. Washing them after cutting destroys vitamins.
- Eat seasonal fruits and vegetables.
- Whole pulses are more nutritious than husked pulses.
- Cook vegetables in iron utensils. This increase the amount of iron.
- There is more iron in jaggery as compared to sugar.
- Special attention should be paid to cleanliness while cooking, serving and eating food. This will prevent many diseases.

Let's see, what you have learnt 12.2

I. Give everyone food as per his/her requirement:



- Balanced food containing calcium and iron.



- Food which gives strength and which contains substances to prevent disease.



- Milk and protein.



- Food containing more iron.

II. Put the sign of correct (✓) or wrong (✗)

- Mother's milk is a child's complete food. ( )
- Unpolished rice contains less vitamins. ( )
- Wash vegetables before cutting. ( )
- Jaggery contains more iron than sugar. ( )

12.3 Storage of food items

Seasonal fruits and vegetables are cheap. Storing them can make us enjoy them in non seasonal time also. Correct storage can also save cooked food for many days from becoming stale and rotten. Food items rot and get spoilt because of fungus or bacteria. Proper storage can save food from fungus and rotting, sometimes for some days and sometimes for some months.

These days packets of chips and salted mixtures, grinded spices, bottled pickles/jams and marmalades are also available in small shops. People like to use such things. This is called preservation of fruits and vegetables. Some farmers preserve things in this way and get maximum benefit of their crops.



Preserved food items in shops, pickles

In homes the following methods are adopted for storage of food items:

1. **Drying:** Preserving things by drying them is quite an old method. Things are dried and the moisture is reduced to such a low level that they are not affected by fungus and they do not rot or decay. Another advantage of drying is that the thing becomes small and light. In this way it is easy to store it. We use dry peas, cereals, pulses, spices etc. every day. Raisins, plums, figs, apples etc. are also dried and stored.

The cheapest and easiest way to dry things is by sun rays. Some things are slightly cooked before storing, like papad, badiyan, macaroni, potato chips etc.

The system of drying and storing meat and fish is quite old. Before drying meat, fish and vegetables, they are salted.

2. **Refrigerating:** Fungus and bacteria do not grow in the cold. Therefore things can be cooled and preserved for a long time. Things are cooled in a fridge or ice box. In the fridge, cooked food, raw vegetables, fruits, butter, milk, meat, fish etc. can be preserved for quite some time.



Fridge

Farmers preserve fruits and vegetables in a cold storage.

In villages people keep food cool in mud utensils with the help of cloth and sand.



Pickle made in oil and spices

3. **With oil and spices:** Salt and sugar are mixed in oil and spices to make a solution in which fungus and bacteria do not grow. In this solution the taste of food is also preserved. In pickles the top layer of oil helps in preventing fungus and bacteria from reaching the pickle.

4. **Use of chemicals:** This is a very technical way. Any excess or lack of chemicals can spoil food. This can adversely affect the health of the person eating the food. Hence chemicals should be used with caution.

At domestic level, we can preserve vegetables and fruits like amla, green mangoes, meat and fish etc. by salting them.

Sugar also works in the same way. Jams and jellies are preserved in sugar. Many items are preserved in vinegar and lemon juice also.

5. **Removing bacteria by cooking:** Cooking prevents the birth of bacteria. By cooking food at very high temperature, birth of bacteria is stopped. Any deficiency in cooking can cause the food to become poisonous. A person eating such food can have diarrhea or vomiting. This is called food poisoning.

6. **Freezing:** If food is cooled and frozen, then fungus and bacteria do not grow on it. But frozen food once melted should not be frozen again.

7. **Canning:** In this method of storage the nutrients in food are not destroyed. Things like fruits, peas, tomatoes are boiled in water and then canned. Things are canned in such a way that air does not reach them.

8. **Fermentation:** This is the oldest method of preserving food. Wine and vinegar are stored in this way.



Canned food items

Let's see, what you have learnt 12.3

I. What is the advantage of removing moisture from food?

_____

_____

II. Mention four methods of storing food.

_____

_____

_____

_____

III. Write the method of storing the following:

Name of things	Method of storing
1. Papad, Badiyan, Macaroni	
2. Butter, milk	
3. Wine, vinegar	
4. Pickle	

Let's Revise

- Deficiency of nutrients can lead to many diseases.
- Deficiency of iron can cause anemia disease.

- Kwashiorkor is caused by the deficiency of protein in children.
- Malnutrition causes rickets.
- Night blindness is caused due to deficiency of Vitamin 'A'.
- Bones of children who do not get enough milk and sun light become deformed. This is called rickets. Editor check
- Vitamin deficiency causes many skin diseases.
- Deficiency of iodine causes goitre. If a pregnant woman does not eat salt containing iodine, then her child can suffer from mental weakness.
- Beri-Beri is caused by deficiency of Vitamin 'B'.
- With some intelligence cheap but tasty nutritious food can be prepared.
- Growing children, adolescents particularly girls, pregnant and lactating women and patients require special nutrients in their food.
- Proper storage can save food items, sometimes for some days and sometimes for some months from becoming stale and rotten.
- Some methods of storing food at home are:

1. Drying
2. Refrigerating
3. With oil and spices
4. Using chemicals
5. Removing bacteria by cooking at high temperature
6. Freezing
7. Canning
8. Fermenting



Exercise

I. How can we prevent anemia disease?

_____

II. What are the symptoms of rickets?

_____

III. Tick the correct answer (✓)

i. Prevention and treatment of rickets-

- Darkness
- Sun light
- Vitamin 'C'

ii. Cause of goitre is-

- Deficiency of iodine
- Deficiency of salt
- Deficiency of water

iii. Deficiency of Vitamin 'B' causes -

- Night blindness
- Kwashiorkor
- Beri-Beri

IV. Select the correct word and fill in the blanks:

i. During adolescence girls need ..... in large amount. (Iodine/Iron)

ii. A child of over six months should be given ..... with milk. (Top food/Salt)

iii. A pregnant woman should eat bellyful of ..... .
(Balanced food/mixture of rice and pulses)

iv. By fermentation, food items can be ..... . (Rotten/preserved)

V. Mark as correct (✓) or wrong (✗)

i. Finger millet and pearl millet have excess iodine. ( )

ii. It is better to have mixed cereals as compared to only one cereal. ( )

iii. With some intelligence inexpensive and nutritious food can be prepared. ( )

iv. Cook rice and vegetables in plenty of water. ( )

v. Food items can be refrigerated and preserved for many days. ( )

VI. What is the meaning of bacteria free?

_____

VII. Write in detail about any one method of storing food.

_____

_____

_____

_____

ANSWERS

Let's see, what you have learnt

12.1 I. Pale transparent skin, yellow gums

II.

Disease	Cause
Rickets	No sunlight on the body
Beri Beri	Deficiency of Vitamin 'A'
Night blindness	Deficiency of Vitamin 'B'

III. Goitre and mental slowness

IV. Eat food containing Vitamin 'A' like green leafy vegetables, yellow and red fruits and vegetables.

12.2 I. Babies and Children - Milk and proteins

Adolescent girl - Food with excess iron

Pregnant woman - Balanced food with calcium and iron

Sick person - food which gives strength and has nutrients to prevent diseases

II.
- ✓
- ✘
- ✓
- ✓

12.3 I. If moisture is removed from things then fungus does not come on them and they do not become stale and rotten.

II. i. Drying ii. Refrigerating

iii. Canning iv. Removing bacteria by cooking

III.

	Name of things	Method of storing
i.	Papad, Badiyan, Macaroni	Drying
ii.	Butter, milk	Refrigerating
iii.	Wine, vinegar	Fermenting
iv.	Pickle	With oil and spices

Exercise

I. Eat food which contains iron like pearl millet, finger millet, jiggery, spinach, meat, fish.

II. Loss of weight, face becomes like that of an old person, protruding stomach, only skin can be seen on bones in the body.

III. i. sunlight

ii. deficiency of iodine

iii. Beri Beri

IV. i. iron

ii. top food

iii. balanced food

iv. preserved

V. i. ✘

ii. ✓

iii. ✓

iv. ✘

v. ✓

Assessment Sheet 4

From Lesson 10 to 12

13

Internal Organs of the Human Body



We will learn in this lesson

- What are the main internal organs of human body?
- What are the functions of these internal organs?

Our body is ours. Hence we have to learn about it, like, how do we breathe? How does the blood flow in the body? How is food digested? How do we think or understand?

Our body is like a machine - like the parts and components of a machine, our body also has many parts. These parts help us to walk, breathe, digest our food, give birth to children etc. these parts are hidden inside our head, chest and stomach. In this lesson we will learn about some of these important internal parts.

13.1 Bones and Muscles

Bones and Muscles help us in walking and in our movements.

Skeleton (Structure of the Bones)

Our Skeleton is in our body. We cannot see it, but just by touching our body we can assume how our bones will be. Some bones are small, some big. Some are flat while others are round. Just imagine

what the body would be like if it did not have any bones. Bones give shape to our body. In a human body there are altogether 206 bones. The longest bone is the thigh bone and the smallest bone is behind the ear drum.

Main functions of the Skeleton

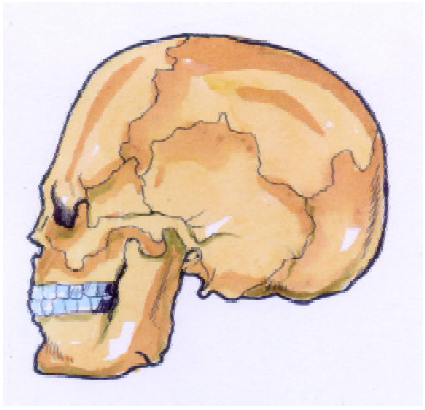
- It gives shape and strength to the body.
- It protects the soft and delicate parts of the body.
- Muscles cover the bones. It is because of our bones and muscles that we are able to walk and move our hands, feet and fingers.

Main Parts of the Skeleton



Spine

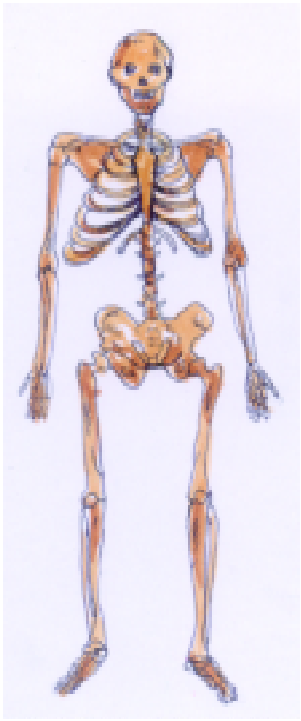
Head: Head is a box made of strong bones. Our brain is safe inside it. Brain is called mind also. Bones of the mouth and jaws are attached to the head. For our eyes, nose etc. there are hollows.



Head

Spine: From the head to the waist, in the middle of back, is our spine. The spine is like a cord of 33 big and small bones. It is for this reason that it has a lot of flexibility. Each bone of the spine is like a band which has a small role in it. These bones are interlinked, one on top of another and form a stem. In this stem the spinal cord is safe.

The spinal cord starts from the back side of the brain and is like a long, thin rope. The spinal cord sends all messages from the brain to different parts of the body.



Skeleton

Ribs: The ribs are like a cage of bones. In this cage the lungs and heart lie safe and secure. In the chest, on both sides are twelve - ribs each. Out of this, on both sides ten - ribs each are attached to the spine.

Shoulder and Hip Bones: These bones help in the movement of hands and legs.

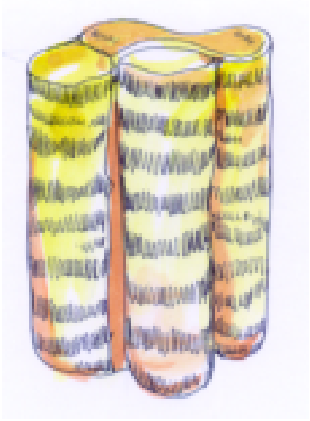
Arm and Leg Bones: These bones give shape to our hands and feet.

Joints: The area where one bone attaches to another bone is called a joint. Some joints can move and shake. They are called dynamic joints like knee, elbow, wrist and fingers. The bones of some joints are static like head joint.

Muscles: Muscles join the bones to one another. They are like flexible strips which help the bones to move easily. There are more than 600 muscles in our body. A special quality of the muscles is that they can become tight or loose. Delicate parts like our heart and our intestines have some special muscles which on their own are able to contract and expand.

These muscles are of two types:

1. **Voluntary Muscles:** These muscles work according to our wishes like muscles that move hand, feet, wrist or fingers.
2. **Involuntary Muscles:** These muscles do not function according to our wishes. They work on their own, like muscles of the intestines push the food down on their own. The muscles of the heart pump blood by themselves.



Voluntary Muscles



Involuntary Muscles

Let's see, what you have learnt 13.1

I. Which internal parts help in the movement of the body?

_____

II. How many types of muscles are there in our body? Write their names?

_____

III. Give two examples of joints that can move?

_____

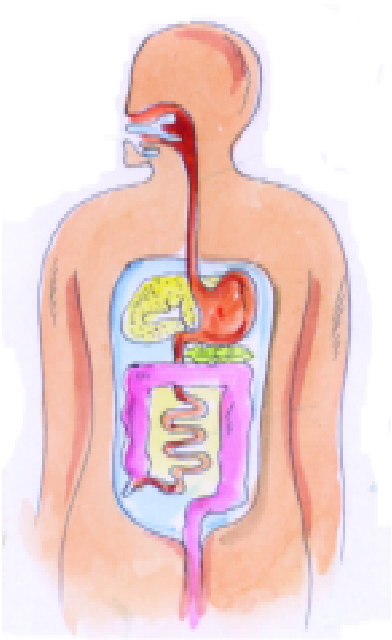
IV. Which organ of our body is safe inside our head?

_____

13.2 Digestive System

Eating food, digesting it and taking the waste food out of our body is the work of digestive system. The main parts of the digestive system are:

Mouth: The mouth has teeth and tongue in it. Adults have 32 teeth in their mouth. As soon as we put food in our mouth, it is chewed by the teeth and broken into small pieces. The mouth has saliva glands from which the saliva comes out. When the teeth chew the food, saliva mixes with the food. It makes the food soft and tender.



Digestive

Food Pipe: This soft food is pushed by the tongue down a pipe. This pipe is called the food pipe. The food goes through the food pipe and reaches the stomach.

Stomach: The stomach is in the abdomen. It looks like a bag. When the food reaches the stomach then the walls of the stomach release some juices. These juices help to digest the food end destroy any pathogens that are in the food.

Small intestine: Food that is digested in the stomach reaches the small intestine. Small intestine is nearly a 7 metres long and narrow tube. Major portion of the food digested here becomes food like a diluted mixture. There is another peculiar thing about the small intestine. Its walls have finger like protrusions in which thousands of fine blood vessels are present. These blood vessels suck all the beneficial substances from the food.

Big Intestine: The undigested and rejected food goes from the small intestine to the big intestine. Big intestine is quite thick. The water mixed in food is sucked over here.

Rectum: From the big intestine the undigested portion of the food reaches the rectum. The undigested part is called faeces, which collects in the rectum.



Liver

Anus: The last part of the rectum is called anus. It is also called anal hole. The useless undigested food goes out of the body in the form of (faeces/ latrine).

Liver: Liver also known as zigar or kaleja in hindi is the biggest gland in the body. Liver is on the right hand side of the stomach in the abdomen. Bile is formed here and it collects in the bile duct. This bile goes into the small intestines through a thin duct. Bile helps in the digestion of food. Apart from making bile, liver also performs many other functions.

Let's see, what you have learnt 13.2

I. Put a sign True (✓) or False (✗) in these statements.

- From eating food to digesting, it is the function of the digestive system. ( )
- A mouth has 42 teeth. ( )
- Most of the food is digested in the stomach. ( )
- Liver is the biggest gland of the body. ( )
- Our stomach is like a bag. ( )

II. Write the names of different parts of the digestive system starting with the mouth.

_____

III. What is the main function of the liver?

_____

13.3 Lungs

Breathing is necessary for every living being. If we do not breathe we start suffocating. When breathing is so necessary, then it is important that we learn the process of breathing. Come, let see, which parts help us in different ways.

- We inhale our breath through the two nostrils of our nose. There are tiny hairs in our nose. They clean air and prevent any dust and particles present in air from being inhaled.
- From the nose the air goes to the larynx. The larynx is like a small box. To speak or for making any kind of sound, the larynx is necessary. The larynx is joined to the trachea on lower side.
- The air goes through the larynx and reaches the trachea. The trachea lies in the middle of our chest and is a straight pipe. The lower portion of the trachea is divided into two tubes. These are called bronchi.
- One tube goes to the left lung and the other goes to the right lung. The air goes through these two bronchi tubes and reaches the lungs.



Lungs

- Our lungs are like balloons. These are inside our chest adjacent to the wall of the chest.
- Inside the lungs the bronchi divide into small pipes. At the end of each pipe these are tiny bags. These bags are called alveoli. The air which we breathe in goes into these bags and expands our lungs.
- Around alveoli there is a network of very fine blood vessels. These pipes suck in the oxygen from the air which is present in the bags.
- Then this oxygenated blood is taken by some blood vessels to the heart.
- The heart sends this oxygen filled blood to different parts of the body. Over here oxygen mixes with the food elements. This action generates energy. When energy is generated in the body, carbon-di-oxide is released. Carbon-di-oxide gas is useless for the body.
- Some vessels carry the carbon-di-oxide blood back to the heart. The heart sends it to the bags present in the lungs.
- When lungs contract and exhale air, the carbon-di-oxide present in the body is also released. In this way everytime we inhale and exhale air, it starts the process of inhaling oxygen and exhaling carbon-di-oxide.
- An adult person inhales air 16-18 times per minute. Exercise, illness and fever etc. increase our rate of breathing.
- Some people who smoke a lot of bidi and cigarette; the alveoli of their lungs contracts. They suffer from respiratory diseases.



Let's see, what you have learnt 13.3

I. From where does the fresh air enter our body?

_____

II. Into how many parts is the lower portion of the trachea divided?

_____

III. Where are the alveoli's?

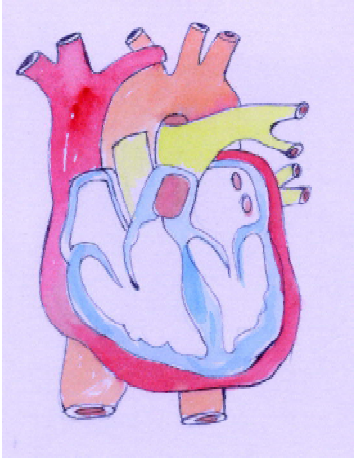
_____

_____

13.4 Heart

Keep your hand to the left side of your chest. You will feel a Dhak-Dhak sound. This sound is due to the beating of the heart. The heart of a healthy person beats 60-80 times per minute. Normally the heart beats 72 times per minute.

- The size of the heart is that of a fist. It is located in a cage made of our ribs, between the two lungs.
- The heart works like a pump. The way a pump sends water to the fields, in the same way the heart sends blood to all corners of our body.
- The heart is made of such muscles which expand and contract throughout our lives. This process of expansion and contraction of the heart is called heartbeats.
- The heart has four chambers, two on the left and two on the right side. The upper left and right chambers of the heart are called atria and the two lower chambers are called ventricles.
- The blood from the left side never meets the blood of the right side.
- The impure blood of our body comes to the right side of the heart. In this blood there is less oxygen and more carbon-di-oxide. The heart sends this blood to the lungs. The lungs purify the blood by removing carbon-di-oxide from it and replacing it with oxygen.



Heart

- The blood with oxygen in it goes to the left side of the heart. Then the heart sends this blood to the whole body.

When the heart beats, the blood from all organs comes to the heart, then from the heart to the lungs, then from the lungs again to the heart and from the heart to all parts of the body. This way the flow of blood is continuous. Regular flow of blood is necessary because it takes various elements of our body from one place to another.

Let's see, what you have learnt 13.4

I. How many chambers are there in the heart?

_____

II. On which side of the heart does the impure blood (full of carbon-di-oxide) come?

_____

III. Where does the heart send the impure blood?

_____

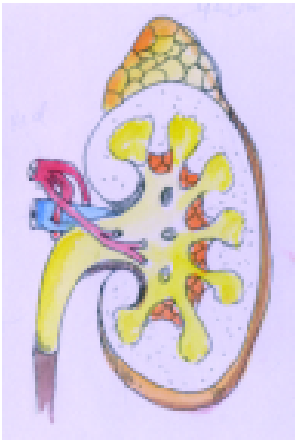
IV. How many times does the heart beat in a healthy person?

_____

13.5 Kidney

In our body some work or the other goes on continuously, like digestion of food, breathing, body growth etc. In this process many harmful substances grow and it is necessary that such waste is removed from our body.

Urine is one such matter or substance. In our urine there are many harmful wastes. To remove them from our body, our kidneys help us. Kidneys maintain a balance between water and mineral salts.



Kidney

- A human being has two kidneys. These are near the waist on both sides of the spine. Their structure is bean shaped.
- Every kidney has a tube which is called ureter. The ureter opens into a urine sac. This sac is called the urinary bladder.
- When the blood reaches the kidney, it is mixed with water and some toxic substances. If there is an increase in their quantity in the blood, then it can be risky.
- Kidney has lakhs of small - small sieves. These sieves filter out the waste from the blood. The clean blood is sent back to the heart and other parts of the body. What remains in the kidney is harmful and waste substances and unusable water. This waste collects in the form of urine and goes from the ureter to the bladder.
- At the bottom of the bladder is a ring of muscles. When we pass urine this ring open out and allows the urine to pass.

Let's see, what you have learnt 13.5

I. How many kidneys are there in the human body?

_____

II. What is the function of the kidneys?

_____

III. Where is the urine collected in the human body?

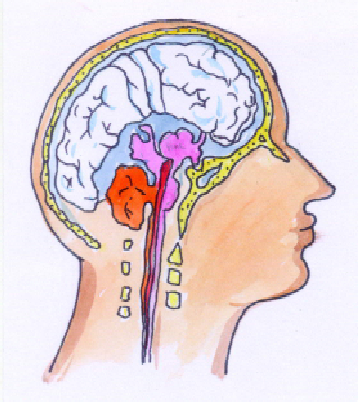
_____

IV. What is the pipe called, which joins the kidney to the bladder.

_____

13.6 Head (Brain, Mind)

It is because of our brain that we think, understand and know what is happening in our body and around us. The brain controls nearly every function of our body. Brain is a soft and fragile organ. It lies secure in the skull. It looks like a walnut whose shell has been broken. Somewhere its colour is pink and somewhere light gray. It is pink because in it and around it there are thousands of thin blood vessels. The oxygen reaches the brain through these vessels. The brain has to work very hard so it needs a lot of oxygen. Different parts of the brain are responsible for different functions.



Head

Based on the functions, the brain can be divided into three parts.

1. A small portion of the brain controls our muscles, like walking, running, lifting our hand etc.
2. The lowest portion of the brain controls those functions which happen on their own, like the beating of the heart, breathing etc.
3. A major portion of the brain handles functions such as speaking, hearing, smelling, thinking, understanding and remembering. This part of the brain is the largest in humans among all creatures. It is because of this that man is able to perform all such functions which other creatures are unable to do.

Let's see, what you have learnt 13.6

I. Where is the brain located?

_____

II. What are the functions of the brain?

_____

_____

III. Into how many parts is the brain divided, according to the work that it performs?

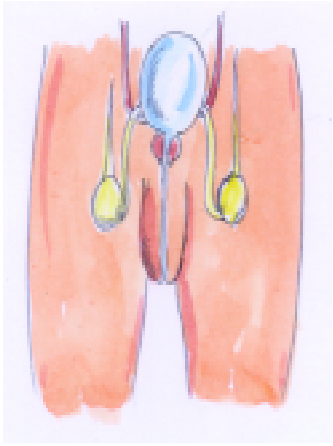
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13.7 Reproductive Organs

All creatures have babies to maintain their species. The organs which help in the birth of offsprings are called the reproductive organs. In human beings two people are necessary to produce children - male and female. There is a lot of difference in the reproductive organs of males and females.

13.7.1 Male Reproductive Organs

Look at the picture and understand the function of each part.



Male Reproductive Organ

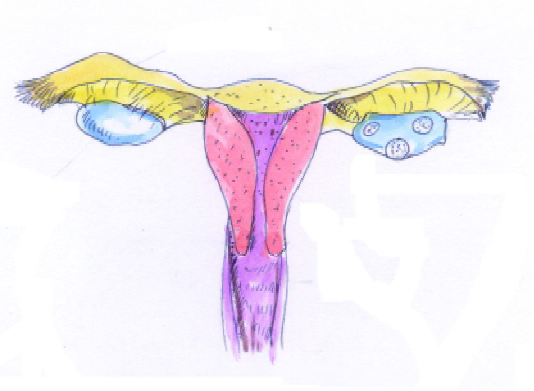
- The male reproductive organ is outside the body.
- Reproductive Organ
- A male person has a pair of testicles.
- Testicles are outside the body and their structure is like a pouch.
- In the testicle the male seeds are formed. These are called sperms.
- From the testicles a tube comes out. The sperms collect in this tube. In this tube a particular fluid is made. When the sperms mix with this fluid, then semen is formed. The semen goes from the penis to the female reproductive organs.

13.7.2 Female Reproductive Organs

Look at the given picture and understand.

- The female reproductive organs are in the lower part of the abdomen, the pelvic area.
- Inside the woman's abdomen is the uterus. The pelvic bones surround the uterus from all sides. They protect the uterus.
- Uterus is very small. It is broad on the top and thin on the lower end. Its walls are made of

thick muscles. The mouth of the uterus is open from the lower end. This is called the cervical opening. This opening is inside the vagina, a little on the upper part.



Female Reproductive Organ

- The vagina open between the thighs and from here, menstruation cycle occurs every month, and menstrual blood leaves the body.
- On the upper side of the uterus are the fallopian tubes. These are thin and around 4 inches long. Their mouth is open and in the shape of a fan.
- Below the mouth of these tubes are the ovaries. In the ovaries are lakhs of tiny eggs cells. From them every month only one mature egg comes out. In a human being only one mature egg comes after every 28 days.
- Mature egg goes from the ovaries straight to mouth of the fallopian tube.
- When the male semen goes into the vagina, lakhs of sperm go quickly into the uterus. From there they go the fallopian tubes. If a mature egg is available then one sperm meets the ovulated egg. The rest of the sperms get destroyed.
- It the sperm burrows into the egg then fertilization takes place and the embryo form of pregnancy is established.
- Fertilized egg enters the uterus and sticks to its wall and the child starts developing from this egg.
- Every month the walls of the uterus thicken waiting for an egg to fertilize. If the egg does not fertilize then all other things ready for the growth of child come out in the form of menstruation.

Let's see, what you have learnt 13.7

I. Why is reproduction necessary?

_____

II. Fill in the blanks by choosing the correct words.

Uterus, Eggs, Sperm, Female

i. In human beings male and ..... reproductive organs are different.

ii. ..... are formed in the ovaries.

iii. ..... are formed in the testicles.

iv. Fertilized egg sticks to the walls of the .....

III. What all is mixed in the semen?

_____

Let's Revise

- The internal organs of our body are hidden inside the stomach, chest and head. These organs are necessary for walking, breathing, digesting food, removing the waste from our body, giving birth etc.
- The Skeletal bones give shape and support to our body. They protect delicate organs of our body.
- The maine parts of the Skeleton are - head, spine, muscles, shoulders, thigh and bones of the hands and legs.
- The main function of the muscles is to give momentum to some delicate part of our bones and body.
- Muscles are of two types - voluntary and involuntary.
- The food is digested in the digestive system. The main parts are - mouth, food pipe, stomach, small intestine, big intestine, liver.
- The lungs help us to breathe. The lungs have lakhs of airbags, where there is a exchange of the gases.
- The function of the heart is to send blood to all parts of the body. Heart is made of very strong muscles.
- The kidneys take out harmful substance in the form of urine from our body.
- The brain lies safe in our head. It controls nearly all functions of our body.
- The male reproductive organ lies outside the body, mainly testicles, sperms, semen and penis.
- Female reproductive organs are in the lower part of the abdomen - mainly, uterus, fallopian tubes, overies and vagina.

Exercise

I. Write the names of four main internal organs?

_____

II. Match the following:

i.	Heart	(a)	Gives shape and structure to the body
ii.	Lungs	(b)	Digestion of food
iii.	Head	(c)	Sending blood to the body
iv.	Kidney	(d)	Think and understand
v.	Skeleton	(e)	Breathing
vi.	Digestion	(f)	Remove harmful waste from the body

III. Tick (✓) the appropriate answer.

i. Lies safe in the head

a. Heart b. Lungs

c. Brain d. Kidney

ii. The main function of the muscle is

a. Give momentum to bones and parts of body which are delicate.

b. Think and understand.

c. Clean the body of waste.

d. Give birth

iii. How many chambers does heart have?

a. Two b. Four

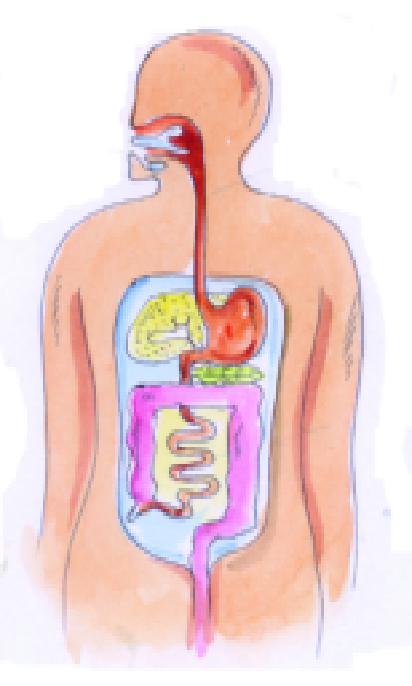
c. Five d. Six

iv. Total number of bones in the body of an adult person is

a. 226 b. 602

c. 206 d. 106

IV. To which part of body does the picture opposite relate?



V. Fill in the blanks:

i. The area where a bone joins another is called a .....

ii. The muscles which do not work as per our desire are called ..... muscles.

iii. Bile is formed in the .....

iv. Air passes through larynx to .....

v. Kidneys have lakhs of small .....

ANSWERS

Let's see, what you have learnt

13.1 I. Bones and muscles.

II. Our body has two types of muscles.

i. Voluntary muscles

ii. Involuntary muscles

III. Joints of knee and elbows.

IV. Brain is safe in skull (head).

13.2 I. i. ✓ ii. ✗ iii. ✗

iv. ✓ v. ✓

II. Mouth, food pipe, stomach, small intestine, big intestine, Rectum, Anus.

III. To forms bile.

13.3 I. Through nose

II. In two parts

III. In lungs

13.4 I. Four II. Towards right

III. In lungs IV. Beats 60 to 80 times

13.5 I. Two

II. Help in throwing out harmful substances (waste) mixed in blood through filteration

III. In the bladder

IV. Ureter

13.6 I. In the skull (head)

II. Brain controls nearly every activity of body. A man thinks, understands and knows as to what is happening around him with the help of brain.

III. Three

13.7 I. To maintain their species

II. i. Female

ii. Eggs

iii. Sperms (male seeds)

iv. Uterous

III. A particular chemical substance (prostatic fluid) and sperms are mixed in semen.

Exercise

I. Heart, brain, lungs, kidneys

II. i. c, ii. e, iii. d, iv. f, v. a, vi. b

III. i. Brain

ii. To give momentum to bones and delicate parts of body

iii. Four

iv. 206

IV. Alimentary Canal (Digestive system)

V. i. Joints

ii. Involuntary muscles

iii. Lever

iv. Cardia

v. Sieves (Nephrons)

14

Common Diseases - their reasons, prevention and protection from them

We will learn in this lesson

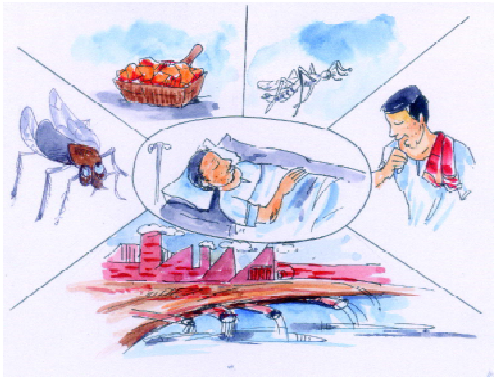
- Infections and non-infectious diseases.
- Symptoms of common diseases, why do they spread and protection from them.
- Reasons for the spread of some non-infectious diseases.
- Steps for protection from diseases.

Seema is suffering from fever for the last three days. Her whole body is aching. She has cough, too. The doctor says; 'she is suffering from flu.' A few days back Seema's daughter, Rinku also suffered from flu. Seema has contacted this disease from her.

This means, one man can contact flu from another person. Are all diseases passed on from one man to another?

We can divide them into two categories from the point of view of their spreading, their prevention and treatment:

(a) **Infectious diseases:** Infectious diseases are those which one man contacts from another. They spread from air, infected food and water. Some diseases are spread by mosquitoes and flies also.



Infectious diseases

Some infectious diseases are spread by small animals such as a fly, mosquito etc. Some diseases are spread by bacteria and virus. One can not see bacteria or virus. We can see them only with the help of a microscope. A microscope enlarged an object and shows us.

(b) Non-infectious diseases: A non-infectious disease does not spread from one person to another. These are many different reasons of non-infectious diseases.

Here belos is the list of some infectious and non-infectious diseases:

Infectious Diseases	Non-infectious diseases
Tuberculosis	Gout
Tetanus, Polio	Heart attack/stroke
Dirrhoea cholera/venereal diseases	Epilepsy
H.I.V./AIDS	Cancer
Contagious boils/wounds	Cataract, Glucoma
Cold, Flue	Aesthma, Allergy
Measles, Small Pox	Abdomenal Ulcer
Itching, Eczema, Irritation	Anaemia
Malaria, Dengue, Elephantiasis	Goitre
Dirrhoea, Dysentery, Typhoid, Jaundice	Cough caused by smoking
Mumps	Diabetes

Let's see, what you have learnt 14.1

I. In how many categories can diseases be divided from the point of view of spreading?

_____

II. Write names of 4 infectious and 4 non-infectious diseases:

Infectious Diseases	Non-infectious Diseases
_____	_____
_____	_____

_____ _____

_____ _____

14.1 Reasons of the spread of some common diseases and proection from them

14.1.1 Cholera



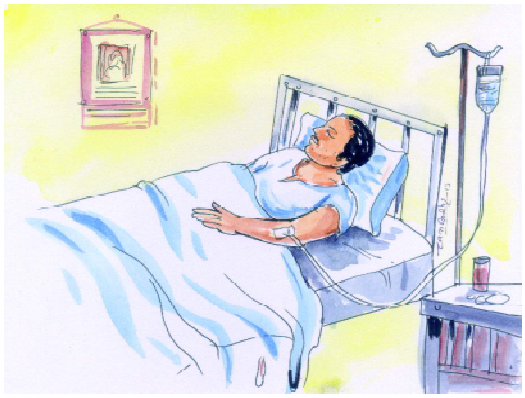
Fly

It is a contagious disease. Generally it spreads in the form of an epidemic. Flies spread cholera.

Symptoms (How to identify it?)

- The patient continuously passes loose, watery stools. He vomits. He may have a mild fever also.
- Due to excessive passing of stool, the patient suffers from deficiency of water. This may lead to his death.

Protection



Cholera

- Cholera spreads through dirty water and infected food. Therefore, one should not eat those food items which are visited by flies. One should not drink dirty water. One should not eat state or rotten food items.
- One should inform the Health Officer, if one finds a person suffering from cholera in his family, village or around.
- The patient should be lodged at a place free from flies.
- One should wash hands will with soap after attending a patient.
- Keep a separate set of utensils for the patient.
- Keep all kinds of food items covered. One should not eat food items sold open (uncovered) in the market.
- The stool and vomit of the patient should be buried in a pit.

14.1.2 Malaria, Dengue, Elephantiasis

All of these diseases are spread by mosquitoes. While sucking blood, mosquitoes suck the bacteria of disease also from the patient's body. When these mosquitoes sting a healthy person, they leave behind these bacteria in that person's blood. This makes a healthy person sick.



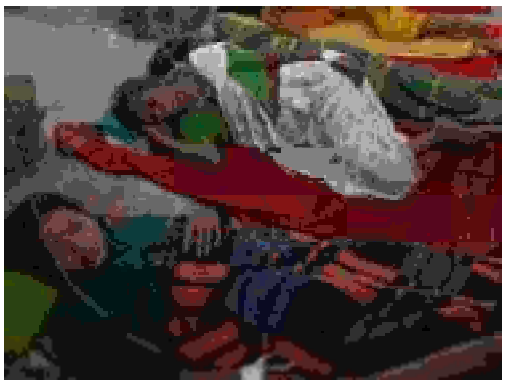
Mosquito

Symptoms (How to identify it?)

- Fever, cold and shivering.
- Head-ache and fever on alternate days.

Protection

- Save yourself from mosquitoes. Use mosquito not co-operate fully with the staff of Malaria Control Deptt.
- If there is fever, get your blood tested. If you suspect; malaria, get the treatment immediately.
- Mosquitoes grow in static water. Donot allow water to accumulate in and around your home.



Malaria

14.1.3 Typhoid

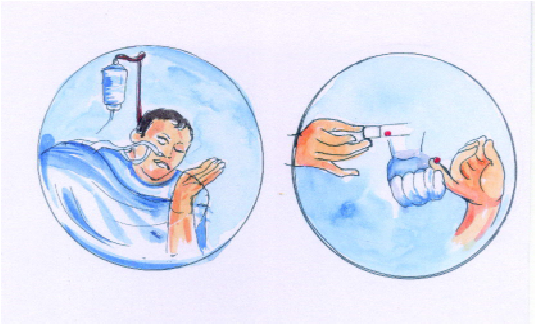
Typhoid is a contagious disease. It is an ailment of intestines. It is caused by contaminated water and food.

Symptoms (How to identify it?)

- There is high fever vomiting, diarrhia and head-ache.
- Fever continuous to rise slowly and slowly.

Protection

- Secure food and water from dirt and fifth. Do not eat food items, cut fruits and vegetables sold open in the market.
- If you see typhoid spreading around, drink boiled water.



Typhoid

- Segregrate the patient from others. Pay special attention to cleanliness of typhoid patient.

14.1.4 Tuberculosis (T.B.)

T.B. generally develops in lungs only. But this disease can develop in any part of the body. T.B. is a highly contagious disease. It develops easily in weak, malnourished children, and persons living with a T.B. patient.



Tuberculosis

Symptoms

- One suffers from continous mild fever.
- One develops cough.
- There is a pain in chest.

Protection

- If someone in the family is suffering from T.B. then the rest of the members in the family should be examined for T.B.
- Children should be vaccinated against T.B.
- Everyone, particularly children, should eat nourishing food.
- While coughing a T.B. patient should cover his mouth with a piece of cloth. One should not spit at all on the floor. Always spit in a spittoon, which has a cover.
- If cough persists for two weeks, if one is losing weight and feeling difficulty in breathing, then one should get oneself immediately examined by a doctor. These may be symptoms of T.B.
- One should get T.B. treatment immediately.
- Treatment of T.B. is rather prolonged one should get one self treated regularly.

14.1.5 Diarrhoea and Dysentery



Diarrhoea and Dysentery

A stool discharge, largely watery is called diarrhoea. If there is phlegm and blood also in stool, it is called dysentery. Diarrhoea may be general and serious as well also. Diarrhoea is dangerous for ill-nourished children. If there is excessive stool discharge, it creates deficiency of water in the body. It is called dehydration. More dehydration can be fatal.

Protection

- Children should be breast-feed instead of bottle

feeding. Mother's milk gives children energy to fight disease. You should not use bottle even if you feed with cow or buffalo milk. It is difficult to keep a bottle clean.

- Child's bed and his surroundings should be maintained neat and tidy. No dirty food (thing) should go into his mouth.
- One can protect one self from practically every disease with the help of nourishing food and good, healthy and clean surroundings.

14.1.6 Cold and Flu



Cold and Flu

Cold and flu are spread by bacteria. These bacteria are called virus also. When a patient sneezes, then the viruses are spread in the air. Then this disease reaches others through air.

Symptoms

- A good lot of sneezing, coughing, cold, running rose and eyes and mild fever.

Protection

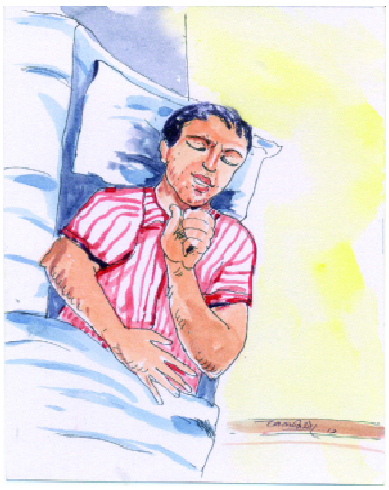
- Good sleep and nourishing food help us in preventing cold and cough. Oranges, tomatoes and fruits rich in Vitamin C are useful.
- People think that getting wet causes cold and cough. But this is not true. Yes, it is true that, if you are already suffering from cold, then getting wet can worsen your condition.
- The patient should place a piece of cloth on his mouth while coughing and sneezing.

14.1.7 Pneumonia

Pneumonia is a dangerous disease of lungs. It can be caused after chicken-pox, whooping cough, flu, aesthma etc. which are respiratory diseases. This disease is highly dangerous for children.

Symptoms

- Breathing becomes quick and fast. There is uneasiness in the chest.
- There is coughing. It is accompanied with yellow, green phlegm also, occasionally phlegm mixed with blood.
- There is pain in the chest and high fever.



Pneumonia

Protection

- Take all these steps, which are taken for protection against cold and flu.
- That person should be looked after well, who faces the danger of pneumonia. Give him nourishing food and take of his rest.

14.1.8 H.I.V./AIDS

AIDS is not a disease in itself. In it the body loses its strength to fight against diseases. In such a situation all kinds of diseases are contacted easily and are not easily cured. AIDS is spread by a virus, called H.I.V. The full form of H.I.V. is Human Immuno Deficiency Virus.



H.I.V./AIDS

- H.I.V. spreads or is transmitted
- When one makes physical relation with on H.I.V. infected person.
- When H.I.V. infected blood is transfused in another person.
- When the needle of an H.I.V. infected person's blood is used for injection to another person, without boiling it.
- To a child born to an H.I.V. infected mother.

Symptoms

There are no separate symptoms of AIDS. But if you see the following symptoms, then consult the doctor immediately.

- If you are suffering from diarrhoea continuously for many days.
- If you are coughing for many days.
- There are, formed in the throat.
- If you carry boils in your mouth for many days.

Protection

- No medicine or treatment helps in AIDS. One can save oneself from AIDS only through awareness and by being careful in one's living.
- Be careful while establishing physical relations, use condom, because H.I.V. is transmitted to the other by semen or fluids of vagina.
- Blood transfusion should not take place to a patient without its proper test and examination.

- One needle should not be used for injection to many persons.
- A child may contact H.I.V. from his mother during pregnancy, delivery or during lactation period, one can save oneself by being careful and through medicines.
- If you suspect AIDS, then get medically, examined. Hospitals carry out this examination free of cost. The hospitals convey the test results only to the person who is tested. His name and test result are not passed on to any one else.

Let's see, what you have learnt 14.2

I. Identify the disease and write its name

Symptom	Disease
The patient is vomitting and passing watery stools.	_____
The patient is shivering and developing fever.	_____
The patient has mild coughing and continuous fever.	_____

II. Suggest two remedies to protect oneself from cholera.

_____

_____

III. Right or wrong

i. Flies spread dengue and malaria diseases. (Right/Wrong)

ii. Blood test is not required if one is running high fever. (Right/Wrong)

iii. Children should be vaccinated for Tuberculosis. (Right/Wrong)

iv/ One should cover one's mouth with a piece of cloth during coughing and sneezing. (Right/Wrong)

14.2 Causes of Non-infectious Diseases

Non-infectious diseases are of many kinds. Some are heriditory. Some are related to life-style and food habits. Some are caused due to growing age and others for many other reasons. Come on, let us know about such diseases.

14.2.1 Heart Stroke

Persons of advanced age in general may suffer from heart stroke. It is more likely to strike persons who eat more greasy food, smoke bidi or cigarette or who have a high blood pressure.

Fat gets deposited in their veins and arteries, which makes their blood flow erratic or it is totally stopped. Due to this the blood does not reach heart. This is called heart stroke.

Symptoms

When heart stroke takes place, the person feels pain in chest. Pain spreads to left shoulder and arm also. The person sweats and feels difficulty in breathing. In such a situation, one consult and take help from a doctor.

Protection

Heart stroke is connected with life-style and food habits. One should eat simple food of less ghee and oil. One should avoid an atmosphere of tension, bidi and cigarette.

14.2.2 Diabetes

Like heart stroke, diabetes, too is a disease connected with life-style. Additionally, diabetes may be hereditory also. Diabetes generally shows itself mostly in 50+ persons. Every person has a general level of sugar in his body. The level of sugar increases in blood in diabetes.

Symptoms

The patient does not know that he is suffering from diabetes. But we can identify it with the following reasons.

- A wound does not heal, or heals very slowly. One has frequent skin problems.
- Weight increases or decreases rather fast.
- Eye sight is blurred.
- There is more of urination.
- Hands and feet dry up.
- Ants throng to such patient's urine.

Protection

- If someone is suffering from diabetes in a family then one should pay attention to food of children right from the beginning. If a child grows too fat and symptoms of diabetes appear, then his blood should be tested.

- The most important thing is to reduce obesity.
- Even if you have a good health, you should get your blood tested every year after crossing 40 years of age.

14.2.3 Cancer

Cancer is a dangerous disease. It can appear on any part of the body. There are many reasons of cancer, such as

- Smoking bidi or cigarette can cause cancer in lungs or lips.
- Chewing tobacco may increase the danger of cancer in the mouth.
- Women generally get breast, uterine and vaginal cancer.



Cancer

Symptoms

- If cancer develops on outer part of the body, then that part develops nodes or knots. Skin may appear reddish.
- Initially, there is no pain. Later on pain starts.
- If cancer has been growing for many days, or it is located on an artery or vein, then it may bleed as well.

Let's see, what you have learnt 14.3

I. Write two reasons of a non-infectious disease.

_____

_____

_____

II. Fill in the blanks:

i. Diabetes is a ..... disease.

ii. There is a pain in ..... when heart stroke takes place.

iii. The wound of a diabetic patient takes a ..... time to heal.

iv. ..... can be caused by chewing tobacco.

14.3 Protection from Diseases

A little protection (precaution) is better than many treatments. If people eat nutritious food, get their children vaccinated and keep their homes and surroundings clean, then they can save themselves from many diseases.

(a) Personal cleanliness or cleanliness of body

- One should wash one's hands with soap after laterine and before taking one's meals.
- One should bathe daily. Bathing protects a man against skin diseases.
- One should brush one's teeth every morning after rising from bed, after meals or after eating sweets.
- Keep children away from domestic animals.
- Use laterine.
- Donot spit on the ground here and there. Please hand, cloth or a handkercheif on mouth while sneezing or coughing.



Wash hands

- Bacteria of disease spread in the air when someone coughs or sneezes. Tuberculosis, chickenpox, measles, flu, cold and cough, pneumonia, diptheria - these diseases spread through air.

(b) Cleanliness in food and drinks

- Always drink clean water. Water should be kept covered in a clean vessel. First boil water and then cool it before giving to very small children to drink. If diseases such as Typhoid, Jaundice, Cholera, Diarrhoea, Dysentery, Polio etc. have spread around, then drink only boiled (and then) cooled water.
- Always keep your food items covered and in a neat and clean surrounding. Donot allow flies, insects etc. to sit on them.
- Donot eat food which has been kept open.
- Always wash fruits before eating them.
- Wash vegetables well before cooking them. Eat only well cooked food.
- Donot eat stale food.

(c) Cleanliness of surroundings (public places)

- Maintain cleanliness around tanks, wells, rivers and taps. Donot defecate or throw garbage near the source of drinking water.
- Have a toilet at home. The government provides grant for it.
- Garbage and waste should be disposed off properly. One should not litter. Mosquitoes and flies breed in garbage and waste and spread many dangerous diseases such as cholera, dysentery, diarrhoea etc.



Throw garbage in dustbin

- Donot allow water to collect in pot-holes. Mosquitoes grow in static water and spread diseases such as malaria, dengue, elephantiasis etc.

(d) Contact with the patient

- Healthy children should sleep separate from the patient.
- Many diseases, such as itching, eczema, tuberculosis, venereal diseases spread if some one touches the patient or wears his clothes or sleeps on his bed. As far as possible, keep such a patient segregated, separate. After nursing and supervising him, one should pay more attention to one's own cleanliness.

(e) Vaccination

Vaccines save children from many dangerous diseases such as Tuberculosis, Tetanus, Diptheria, Whooping cough, Polio, Chicken-pox, Typhoid etc. Children should compulsorily vaccinated for them.


Vaccination

(f) Habits, which affect health adversely

Some habits of men such as drinking and smoking not only harm them, but other persons who live around.

- Drinking (alcohol) leads to many diseases of liver, such as liver sirhosis. Drinking more than required is also the main reason of the problems related to health. Intoxication experienced by the drunkard not only adversely affect his health, it harms other members of the family also.
- Apart from alocohol, other intoxicating things such as tobacco, heroine etc. also affect health adversely. Danger and chances of oral (mouth) cancer increase in the case of a person who chews tobacco.

Come on, let us find out why it is dangerous to smoke bidi, cigarette etc.?

- Smoking increases the danger of cancer in lungs or in lips.
- Smoking causes many respiratory diseases. Smoking can be fatal for persons suffering from aesthma.
- It can cause ulcer in abdomen.
- Chances of suffering from pneumonia increase more in the children of smoking parents in comparison with the children of non-smoking parents. Such children suffer more frequently from respiratory diseases.
- If a pregnant women smokes, it adversely affects her yet to be born child.



Smoking

(g) A good amount of money is spent on alcohol, bidi, cigarette, tobacco etc. If this money is spent on the nutritious food then children and the whole family can have a good health.

Let's see, what you have learnt 14.4

I. Suggest two remedies for personal cleanliness.

_____

_____

II. How do the bacteria of disease reach in the air?

_____

_____

III. Write the names of three diseases spread by dirty water.

_____ _____ _____

IV. What are the diseases against which children are vaccinated?

_____

_____

V. What are two harms of smoking bidi-cigarette?

_____

_____



Let's Revise

We divide diseases into two categories from the point of view of their spreading, prevention and treatment:

1. Infectious 2. Non-infectious

- A person contacts infectious diseases such as Tuberculosis, Cholera, Small pox, Flu, Cold & Cough from other person.
- Non-infectious diseases such as Diabetes, Epilepsy, Glaucoma, Cancer, Aesthma etc. are not contacted by one person from another.
- Some common diseases such as Cholera, Typhoid, Diarrhoea, Dysentery are spread by infected or dirty food and water. One can save oneself from them by maintaining cleanliness.
- Mosquitoes spread Malaria, Dengue, Elephantiasis etc. One can save oneself from them by avoiding mosquitoes.
- Tuberculosis, Cold and Cough, Flu and Pneumonia spread through air. If the patients keep their mouth covered when they cough or sneeze, then the danger of spreading of these diseases can be lessened.
- An AIDS patient appears healthy. But his semen, or watery discharge from a woman's vagina, or through blood, AIDS may spread.
- Heart stroke and Diabetes are non-infectious diseases. One can save oneself from these diseases with a change in his life-style, food and drink etc. These diseases can be hereditory also.
- Cancer is a non-infectious disease. It is caused due to many reasons. Smoking intoxicating things and the use of tobacco increase the danger of cancer.
- One can save oneself from many diseases by eating nutritious food, maintaining personal cleanliness, practising cleanliness in food and drinks and by ensuring cleanliness around.
- Children can be saved through vaccination from many diseases such as Tetanus, Diptheria, Whooping Cough, Polio, Tuberculosis and Chicken-pox.

- Drinking (alcohol), indulging in other intoxication, smoking etc. - such habits not only harm the person concerned but others also who live nearby. By avoiding such habits a person can improve not only his own health, but that of his children and his whole family.

Exercise

I. What is called an infectious disease?

_____

II. Write two reasons of non-infectious diseases.

_____ _____

III. Select and place the following diseases into infectious and non-infectious categories:

Tetanus, Small Pox, Cancer, Diabetes, Flu, Malaria,
AIDS, Heart stroke, Anaemia, Glaucoma

Infectious	**Non-infectious**
Tetanus	Glaucoma
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

IV. Fill in the blanks:

i. Mosquitoes spread ..... disease.

ii. Passing more stools may cause ..... in the body.

iii. Typhoid and ..... spread from eating infected food.

iv. If cough persists for more than 2 weeks continuously, then ..... get a blood test.

v. AIDS is tested ..... in hospitals.

V. Tick right (✓) and wrong (✗) in the following:

i. Smoking can cause cancer of lungs and lips. ( )

ii. People eating simple food can suffer heart stroke. ( )

iii. Quick healing of wound is the symptom of diabetes. ( )

VI. Suggest three remedies of safety against diseases.

_____

_____

_____

VII. Which habits affect health adversely?

_____

_____

_____

Come on, let us practise and see

From which diseases do the people around you suffer? What do they do to save themselves from them?

ANSWERS

Let's see, what you have learnt

14.1 I. Two

II. Infectious Diseases: Tuberculosis, Tetanus, Polio, Cholera etc.

Non-infectious Diseases: Gout, Cancer, Epilepsy, Aesthma etc.

14.2 I. i. Cholera

ii. Malaria, Dengue, Elephantiasis

iii. Tuberculosis

II. One should not eat food items which flies have visited. One should keep food items covered and should not eat things which are sold open in the market.

III. i. Wrong

ii. Wrong

iii. Right

iv. Right

14.3 I. i. Wrong methods of eating and drinking and life-style.

ii. Growing age.

II. i. Connected with life-style

ii. Chest

iii. Late

iv. Cancer

14.4 I. i. Bathe daily

ii. One should wash one's hands with soap before taking food and after laterine.

II. From coughing and sneezing.

III. Typhoid, Jaundice, Diarrhoea.

IV. Tuberculosis, Tetanus, Diptheria, Whooping Cough, Polio, Chicken-Pox

V. i. There can be cancer of lungs and lips.

ii. There can be respiratory diseases.

Exercise

I. Diseases contacted by a person from another are called infectious diseases.

II. Diabetes, Cancer

III.

Infectious	**Non-infectious**
Tetanus	Glaucoma
Small Pox	Cancer

Flu	Diabetes
Malaria	Heart Stroke
AIDS	Anaemia

IV. i. Malaria

ii. Water

iii. Cholera

iv. Test by a doctor

v. Free of cost

V. i. ✓

ii. ✗

iii. ✗

VI. i. Keep your body clean.

ii. Maintain cleanliness in food and drinks.

iii. Cleanliness of surroundings is a must.

VII. Drinking alcohol and smoking bidi, cigarette.

15

Garbage and Waste and its Disposal



We will learn in this lesson

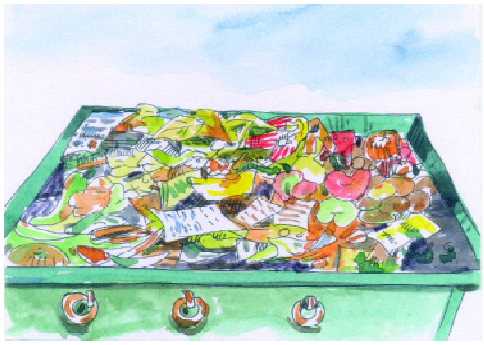
- What is organic and inorganic garbage and waste?
- How to dispose of garbage and waste?
- How can garbage and waste be re-used?
- What is the importance of neatness and cleanliness?
- How to use dustbins properly?
- What harm is caused to environment by garbage and waste?

All of us sweep and clean our home and courtyard daily. In this way a lot of garbage and waste is collected. It has many items which rot. After rotting they mix with the soil. There are many such items also which do not rot. We can make compost fertilizer from rotting items. Items, which donot rot, can be used for other purposes, other things can be made by melting or transforming them. Therefore, it is necessary that different kind of garbage and waste is put into different dustbins. And then it is re-used. In this way, we shall keep our environment also neat and clean.

15.1 Garbage and waste of different kinds

We clean our home and courtyard daily. Some garbage or waste is collected practically everyday. It

has many items or things - dust, paper, leaves which have fallen from plants and trees, old, torn clothes, peels of vegetables, left over food etc. All of them are such things which not in a few days. After rotting they mix in the soil. We call this degradable garbage and waste. We can call it organic waste also.



Organic Waste



Inorganic

Sometimes garbage and waste has nails of iron, pieces of glass, plastic and polythene items also. If we throw these things on the ground, they donot rot over a long period of time. They remain lying on the ground as they were. Plastic and polythene never rot. We call it non-degradable garbage and waste. We can call it inorganic waste also.

Let's see, what you have learnt 15.1

I. Which garbage and waste is called degradable?

_____

II. Which items are included in non-degradable garbage and waste?

_____

15.2 Disposal of Garbage and Waste

What do we do with the garbage and waste which we generate in cleaning? Mostly we throw it here and there as it is. Or at best, we collect it and make a pile. If there is an old well or pond around, we throw it there. If we live by the bank of a river, we throw it in the river. You must have seen the piles of garbage and waste near the roads. It is that garbage and waste which has come out from your and our homes.

Think, is it the right way?



Showing dumping waste in river

If we want, we can make this garbage and waste re-usable.

That the garbage and waste could be re-used, for this we should have a comprehensive understanding of its proper disposal. Disposal means its proper resolution, to put it to a total end.



Throwing organic waste in compost pit

Proper disposal will take place only when different kinds of garbage and waste are segregated and kept aprt. We should keep those things separate which can rot, i.e. those parts of waste which are degradable. We should not mix them with those things which do not rot, i.e. non-degradable items.

We can put in a pit things which rot, such as peels of vegetables, left over food, paper, leaves etc. We can dig such a pit in a corner of our courtyard or our household garden. If required, we may sprinkle some water in this pit periodically.



Re-using iron, waste, scrap

After sometime there would be very good fertilizer in the pit. We can use this compost manure in our flower-pots and home garden.

We should keep separate our non-degradable waste such as iron nails, pieces of glass, plastic, polythene etc. This is lifted and carried away by scrap collectors. Scrap collectors pay some price also for these things in the cities. These things are sent to factories. These things are melted or burnt to manufacture many useful items. In this way, garbage and scrap and waste are re-used.

Let's see, what you have learnt 15.2

I. What use can be made of the garbage and waste of rotting items?

_____

II. How is garbage and waste of non-degradable items used?

_____

_____

15.3 Different kinds of garbage and waste, separate dustbins

How to keep the garbage and waste segregated? The best method for this is to have separate dustbins for different kinds of garbage and waste. One dustbin for organic waste i.e. for things that rot; and a separate dustbin for inorganic waste items.

We can do it in our homes also. These days in big cities sets of 2 dustbins each are provided at different places. On one is written : Organic; on the other : Inorganic. In hospitals, there are generally dustbins of 3 colours: Green, Red and Black. Green dustbin is for organic waste; Red for glass vials, bottles and plastic etc. and Black for rotten things. In many countries, you will find still more categories for waste items. Dustbins, such as one apart for paper, another for cloth, one for peels of vegetables and fruits and left over food, another for items of iron, yet another for glass pieces, and a separate one for plastic. But we can manage by two dustbin in our home.



Two kinds of Dustbins

15.4 Take care of our Environment

It is the responsibility of each one of us that we maintain cleanliness and take care of our environment. We can save our lives and our earth only by keeping our surroundings neat and clean. In our environment are included all things that surround us, whether made by nature or by man. Proper disposal of garbage and waste is required so that our environment remains clean and free from pollution.


Burning waste


Water pollution from factories

Generally we throw our garbage and waste by the roadside or in fields or in parks. It contains many things which render the soil barren. Sometimes we set our garbage and waste to fire. In this tyres of rubber, plastic and polythene bags are burnt. The smoke emanating from them pollutes the air. This polluted air goes inside our body when we breathe. It makes us sick.

Waste, coming out of factories is straight away discharged into rivers quite often. We dump household waste into drains and rivers many times. This, too, has many polluting objects. Garbage and waste dumped in the open flows with rain water and mixes with soil and sources of water. All these factors pollute water. When we drink this polluted water, it gives rise to many diseases. When this water is used for irrigation, then this pollution reaches into our crops also.

15.5 Cleanliness of surroundings

When we step out of our home, we see dirt and filth here and there. It is everywhere: on roads, in bazar, in streets on railway stations and bus stands, in parks, in cinema houses, everywhere. There are peels of peanuts, pieces of paper, empty packets of eatables at some places, whereas there are peels of fruits, empty plastic tumblers and polythene bags at others. There is hardly a spot which appears totally neat and clean. So much so, that we do not see cleanliness in temples either. Have you ever thought : how does this fifth spread? You and I spread it.

We throw waste casually anywhere. Even if the dustbin in two steps away, we donot reach it. We throw waste just where we are sitting.



Spitting everywhere - this is our most common habit. Nothing wrong appears to us in it. You might have seen some persons spitting in buses and trains also. This not only spreads filth, it causes inconvenience to other fellow passengers also. But do these people consider it so?

You must have witnessed another scene also. In cities, people dispose of their household sweepings and waste filled in polythene bags. They throw it any where by the roadside. They donot put it in dustbins. If they reach the dustbins, they throw there casually from a distance. Polythene bag is not placed in the dustbin. Is it O.K.?

When people return from foreign countries, they highly priase the cleanliness they raw in those countries. They say that you will not find even a match stick on the road there. Can we not do this in our country? Can we call ourselves a civilised citizen even when we are spitting and spreading dirt and filth all around?

We will have to make a beginning of cleanliness from our own end. We will have to change our

habits. When all of us cultivate the habit of cleanliness then alone we will be able to make our country spick and span. Then alone we will be able to keep our environment clean. If environment is clean, then alone we will be healthy. For this we should

- not either here and there
- not use polythene
- not burn tyres, plastic, polythene etc.
- make compost manure from organic wastes
- ask our children and younger brothers and sisters to practise cleanliness.

Let's see, what you have learnt 15.3

I. Why are two dustbins kept?

_____

II. How does garbage and waste pollute water?

_____

_____

III. What things are included in environment?

_____

_____

Let's Revise

- The garbage and waste of things that rot is called degradable waste. We can call it organic waste also.
- The garbage and waste of those things which do not rot is called non-degradable waste. It can be called organic waste also.
- Garbage and waste can be made re-usable.
- Compost manure can be made from organic garbage and waste.
- The waste of iron, glass, plastic etc. is melted in the factories and then many things are made out of it.

- We should keep separate dustbins for organic waste and inorganic waste.
- Environment is also polluted by garbage and waste.
- If environment is clean, then we shall be healthy.
- We should not throw garbage and waste here and there. We should throw it in dustbin only.

Exercise

I. Write down the answers of questions:

i. What should we do for proper disposal of waste?

_____

ii. How will you make compost manure from waste?

_____

iii. How many types of dustbins should we keep at homes?

_____

iv. Which waste is thrown in Green Dustbin in hospitals?

_____

v. How can we keep our environment neat and clean?

_____

_____

II. Tick right (✓) and wrong (✗).

i. We can make compost manure from rotten things. ( )

ii. Waste lying in the open does not pollute environment. ( )

iii. Organic and inorganic waste should be kept separate. ( )

iv. The waste coming out of factories pollutes rivers. ( )

v. Proper disposal of waste means 'Dump it in the open'. ( )

III. Out of the following things, which are degradable, and which are non-degradable.

Dry leaves, polythene bags, green grass, plastic buttons, shreds of cloth, broken electrical switches, pieces of card-board, old newspapers, glass bottles.

Degradable	Non-degradable
_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

IV. Complete the sentences, by putting correct word in blank space.

i. We make compost manure using ..... waste. (organic/inorganic)

ii. Smoke of burning plastic ..... air. (pollute/scent)

iii. Flowing in the rain water, waste mixes with ..... (sources of water/air)

iv. Polythene makes the soil ..... (barren/fertilized)

V. What can we do to keep the environment neat and clean? Write it in 6-7 sentences.

_____

_____

_____

_____

_____

_____

_____

_____

Come, Let us practice and see

- Dig a pit for compost manure in your home garden. Make fertilizer in it from waste.
- Place two dustbins at home. Write 'ORGANIC' on one, and 'INORGANIC' on the other. Put your waste in them correctly as per writing.

ANSWERS

Let's see, what you have learnt

15.1 I. Such garbage and waste which rots quickly and mixes with the soil.

II. Iron nails, pieces of glass, plastic, polythene etc., which do not rot for a long time.

15.2 I. One can make compost manure from it.

II. It is melted or burnt in the factories and many useful things are made from them.

15.3 I. One for organic waste, the other for inorganic waste.

II. In factory wastes there are some chemicals also. The garbage and waste of households is thrown into rivers and drains many times. This pollutes water.

III. Environment includes all things around us, whether they are made by nature or man.

Practice

1. i. Organise and inorganic waste should be kept separate.

ii. We will throw in a pit things which rot. If required, we will sprinkle water on them off and on.

iii. Of two types.

iv. Organic waste.

v. Through proper disposal of garbage and waste.

2. i. ✓ ii. ✖ iii. ✓ iv. ✓ v. ✖

3.

Degradable	**Non-degradable**
Dry leaves	Polythene bags
Green grass	Plastic buttons
Shreds of cloth	Broken electrical switches
Pieces of card-board	Needle
Old newspaper	Glass bottle

4. i. Organic

ii. Polluted

iii. Sources of water

iv. Barren

Assessment Sheet 5

From Lesson 13 to 15